

Master's Program in Quantitative Finance (QFin)

Advanced Topic in Banking



Univ.Prof. Dr. Rainer Jankowitsch

- Economic function of banks and regulatory environment
- Principles of bank-wide risk management (ICAAP)
- Credit risk measurement and validation
- Credit risk management
- ICAAP analysis

- Required reading:
 - ECB Guide to the internal capital adequacy assessment process (ICAAP), November 2018
- Additional literature:
 - Heffernan, S.: Modern Banking (2005), ISBN: 978-0470095003
 - Koch, T. W., MacDonald S. S.: Bank Management (2005), ISBN: 978-0324289275
- Regulatory documents
 - OeNB/FMA: Guidelines on Bank-wide Risk Management. Internal Capital Adequacy Assessment Process
 - OeNB/FMA: Guidelines on Credit Risk Management – Rating Models and Validation
- Legal acts:
 - Austrian Banking Act (BWG)
 - Capital Requirements Regulation (CRR)
 - Capital Requirements Directive (CRD)

Warm-Up Case

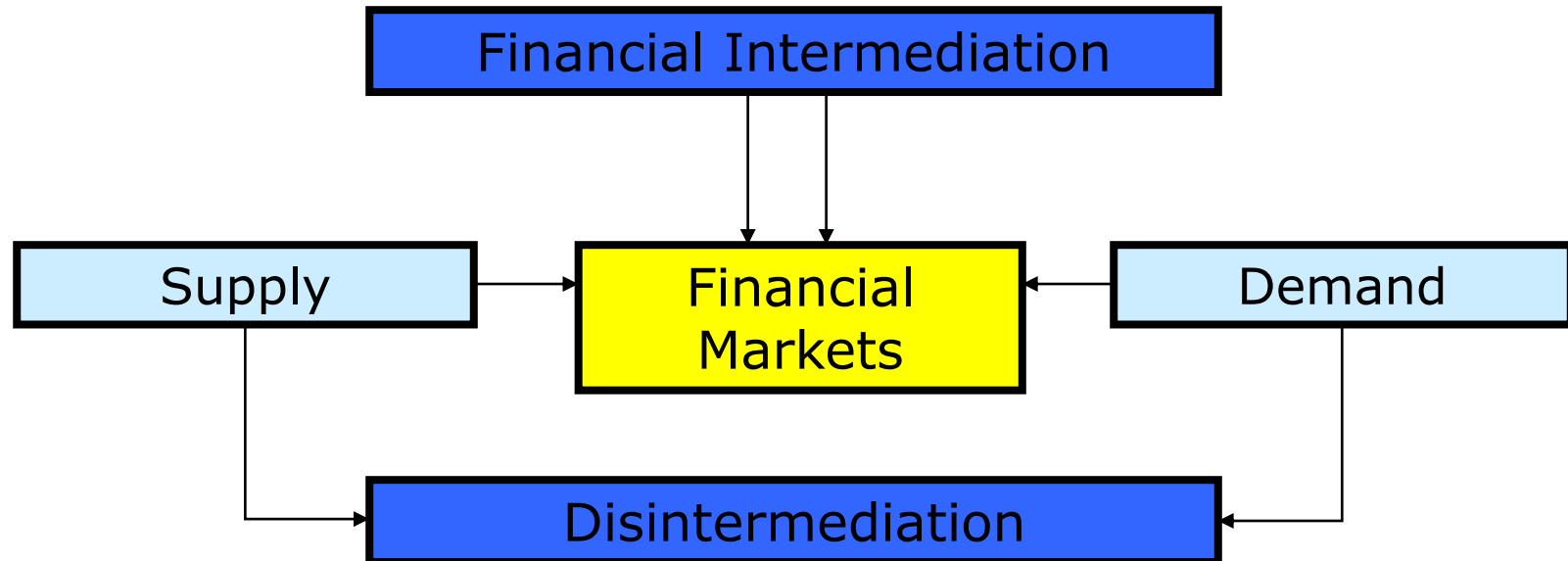
Economic Function and Regulatory Environment



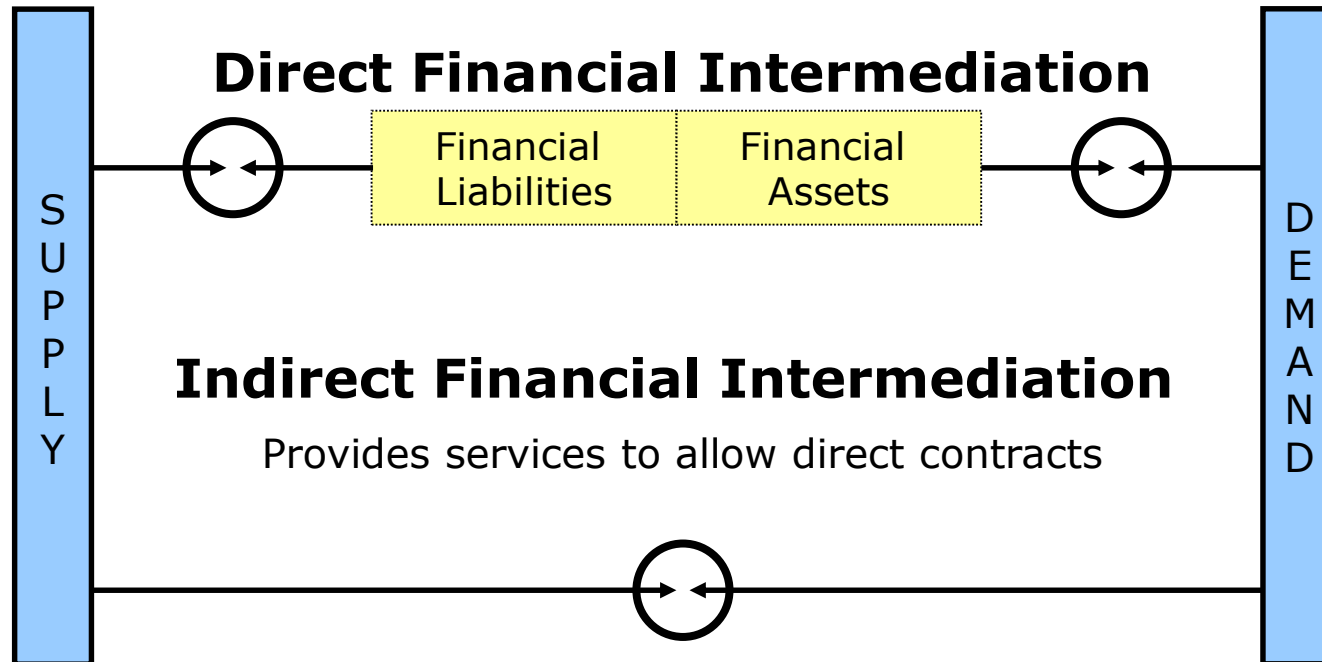
- = “Any place where supply and demand for financial securities/contracts are matched.”
- We can differentiate across five economic agents (Oesterreichische Nationalbank – OeNB):
 - Financial corporations (banks)
 - Nonfinancial corporations (corporations)
 - Government
 - Private households
 - Rest of the world

- Who is supplying and demanding money?

- Thus, the financial systems covers financial markets, financial intermediaries and any other institutions that allow households, corporations and governments to implement their financial decisions.



- Financial intermediaries:



- Direct financial intermediation:
 - Act as legal counterparty
 - Accumulate financial liabilities and assets
 - Take over risks (e.g. credit risk)
 - Have to management their positions
 - Examples: Banks, insurance companies, pension funds, ...
- Indirect financial intermediation:
 - Provide only services:
 - Gather information
 - Matching service
 - Receive fees
 - Examples: Rating agencies, exchanges, ...

- Banks provide important transformation functions as financial intermediaries.
- Transformation functions:
 - Lot size transformation
 - Maturity transformation
 - Risk transformation
 - Information transformation
- Additional function:
 - Transaction banking (processing and settlement of payments)

- Lot size transformation:
 - Borrowers and lenders often need to finance/invest different notional amounts (lot size).
 - Borrowers need large amounts to finance their projects.
 - Lenders are often retail clients and invest small amounts.
 - Therefore, it is necessary to bundle small-size savings.
 - This transformation function is performed by banks by taking in small-size amounts and granting large loans.

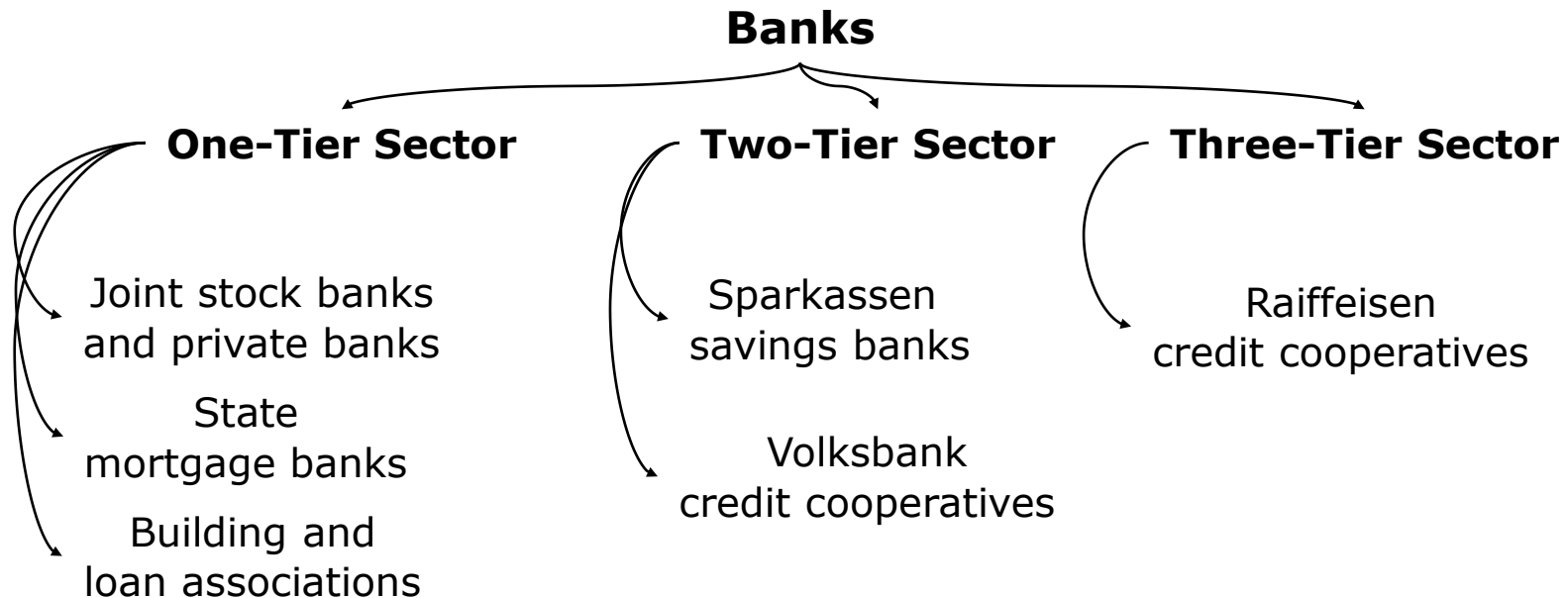
- Maturity transformation:
 - Borrowers and lenders often have different investment horizons, i.e., need funds for different maturities.
 - Borrowers often need long-term financing for their projects.
 - Lenders, especially retail clients, often have short-term horizons (e.g., need to withdraw on a daily basis).
 - Thus, borrowers are facing roll-over risk.
 - This maturity risk is managed by the bank by taking in short-term funds and granting long-term loans.

- Risk transformation:
 - Lenders face the risk that borrowers do not fulfill their contractual obligations, e.g. miss payments or default (credit risk).
 - This risk can be mitigated by diversification.
 - Banks allow their clients to diversify small amounts by (indirectly) investing into diversified portfolios.
 - Otherwise a small lender would be exposed to the credit risk of one individual borrower.

- Information transformation:
 - Lenders need to evaluate the credit risk of borrowers.
 - Lenders are often confronted with asymmetric information: Borrowers know more about their credit risk than the lenders.
 - It is a costly process to gather the necessary information to overcome this asymmetric information problem.
 - In addition, a costly monitoring is necessary to evaluate the fulfillment of contractual obligations.
 - Banks take over these processes.

- Bank-wide risk management:
 - Providing all these transformation functions implies that banks have to take on multiple risks and manage these risks:
 - Credit risk
 - Market risk (interest rate risk, foreign exchange risk, ...)
 - Operational risk
 - Legal risk
 - ...
 - **The quantification and management of these risks is the central task of the banking business.**

Types of Banks in Austria



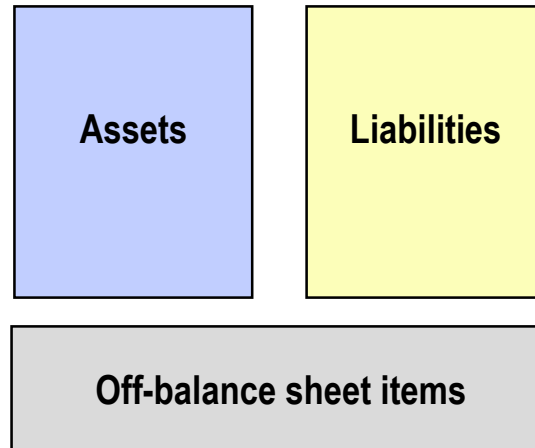
- The Austrian Banking Act defines banks in Article 1. (1):
 - “The term "credit institution" refers to an institution authorised to carry out banking transactions on the basis of Article 4 or Article 103 no. 5 of this federal act, or on the basis of special provisions under Austrian federal law. Banking transactions include the following activities if carried out for commercial purposes:”
 - Important examples of activities:
 - Deposit business
 - Current account business
 - Lending business
 - Custody business
 - Trading business
 - Securities underwriting business

- The Austrian Banking Act defines banks in Article 1. (2):
 - “The term "financial institution" refers to an institution which is not a credit institution as defined under para. 1 and which is authorised to conduct one or more of the following activities for commercial purposes if they are conducted as the institution's main activities:”
 - Important examples of activities:
 - Leasing business
 - Advice to undertakings on capital structure (including M&A)
 - Credit reporting services
 - Safe deposit services
 - Certain payment services

- Why do banks exist?
 - In a frictionless market there is no need for banks.
 - In the academic literature there are two main explanations why banks exist:
 - **Transaction costs:** Small transactions are very costly. Banks allow to reduce these costs by economies of scale (see Santomero [1984] for a review).
 - **Asymmetric information:** Borrowers know more about their economic situation. This asymmetry can be reduced by bearing a cost. These costs can be reduced by economies of scale, i.e., only the bank invests these costs and not every individual lender (see Diamond [1984]).

Balance Sheet and Income Statement

- Balance sheet positions:



Balance Sheet and Income Statement

- Income Statement:

Profit	Loss
Interest income	Interest expenses
Fee and commission income	Fee and commission expenses
Profit from trading	Loss from trading
Other income	Other expenses
	Risk provisions
	Personnel expenses
	Depreciation and amortization
	Taxes

Balance Sheet and Income Statement

- The balance sheet and income statement allow a first evaluation of the economic situation of a bank.
- However, it is not possible to evaluate the risks of the bank from this information source.
- This is especially relevant for banks using derivative and structured products.
- This is why regulation is more focused on assessing these risks with additional models and enforcing disclosure requirements.

- The business model is the strategic choice of a bank.
- An important aspect of the business model is the funding mix:
 - Retail-funding
 - Wholesale-funding
 - Capital markets-orientation
- The key aspects of the business model are asset side activities:
 - Loan portfolio
 - Trading/Investment portfolio
 - Fees from underwriting, M&A, market maker activities
 - Fees from payment and settlement services

- For a typical bank (e.g., Austrian medium-sized bank), the loan portfolio will be the most important asset class.
- Investments into bonds and stock can be part of a strategic asset allocation.
- These capital market investments together with short-term deposits (placing money with other banks or with the central bank) can also be used as liquidity buffer.
- The loan portfolio can be focused on different characteristics:
 - Short-term vs. long-term products
 - Different customer base (retail, corporates, projects, sovereigns)
 - Different collateral
 - Fixed-rate vs. adjustable-rate
 - Term vs. revolving loans
- These characteristics will determine the risk profile (credit risk, market risk, liquidity risk, operational risk, ...).

- Typical products are
 - Mortgage
 - Auto loan
 - Consumer credit
 - Overdraft limit
 - Business term loan
 - Revolving loan / line of credit
 - Accounts receivable financing / factoring
 - Commercial mortgage

- The key to successful business models is an accurate risk-based loan pricing and a sound loan granting process, aligned with the capital and liquidity budgeting.
- Within the loan granting process rating models are employed to estimate central credit risk parameter:
 - Probability of Default (PD): Measures the probability to default on the loan contract in a one year time window.
 - Loss Given Default (LGD): Measures the expected economic loss in a default relative to the loan exposure.
- Together with the Exposure at Default (EAD) these parameters determine the credit risk of a loan contract.
- PD and LGD estimation requires a sound data collection process and state-of-the-art statistical tools.

- Banks use different funding sources beside equity:
 - Retail deposits
 - Wholesale funding by non-banks
 - ECB funding
 - Money market deposits
 - Bonds
- Retail deposits are considered as stable funding source, in particular, because deposit insurance systems protect these deposits.
- Wholesale funding are raised from non-natural person and are not protected and, thus, seen as a risky funding source.

- ECB funding against eligible collateral consists of
 - Main refinancing operations: Fixed rate tenders
 - Marginal lending facility
- All other funding sources make use of the financial markets and are linked to standardized financial instruments.
- These instruments can be used for particular aims
 - Short-term liquidity management (money markets)
 - Long-term stable funding (bond markets)
- Other financial instruments allow to manage / hedge the underlying risks in the portfolio, e.g.:
 - Swap markets
 - Option markets

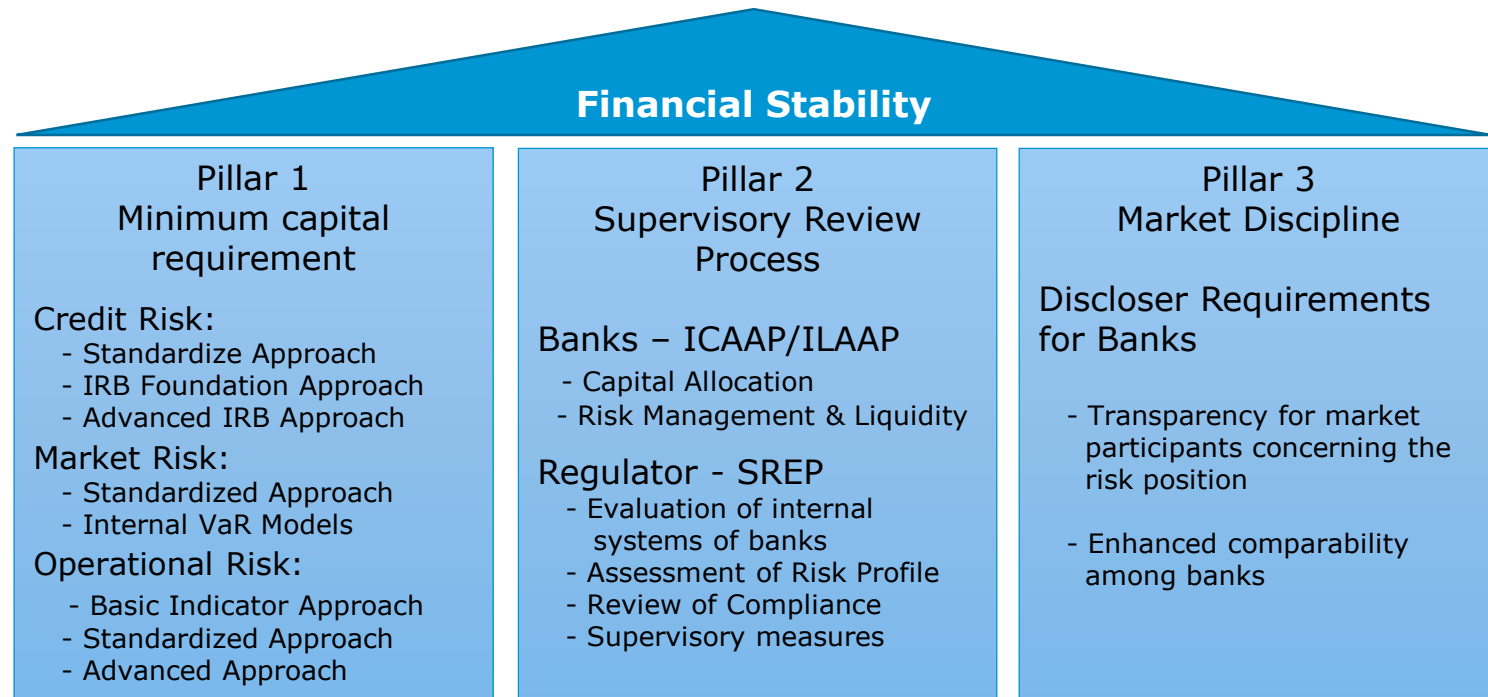
- Theory of regulation:
 - Banking crises have serious consequences for their clients and the entire financial system (e.g., consequences of the financial crisis).
 - In order to prevent greater losses, it is often necessary for governments to save banks (i.e., provide guarantees or directly inject money) - "too big to fail".
 - Therefore, banks are regulated in order to keep the probability of crises small and to minimize the potential costs.
 - Regulation is thus intended to limit the risks that banks take within the scope of their functions.
 - The regulations make (in simplified terms) a minimum level of capital in relation to the risks taken mandatory.
 - There are also regulations on deposit protection.
 - Regulation thus intervenes directly in the market. Possible inefficiencies and regulatory arbitrage must be considered.

- Reasons for regulating banks:
 - Great influence on the money supply in the economy.
 - Due to the possibility of economies of scale (which is one of the reasons for the existence of banks) there is a tendency towards monopoly.
 - Banks are strongly interconnected (e.g. by obtaining liquidity on the money market or by derivatives). Therefore there is a high systemic risk (i.e., possible domino effects).
 - Without regulation, investors would withdraw their money in the event of small changes in a bank's earnings situation (bank run).
 - Retail clients should be protected from asset losses due to ignorance and inexperience.
 - For an overview of the academic literature see Bhattacharya et al. [1998].

- National and international financial supervision and regulation:
 - At the national level, the **Austrian Financial Market Authority** (FMA) and the **Oesterreichische Nationalbank** (OeNB) are responsible for the supervision of banks.
 - Since there are strong international networks of banks, an international harmonization of banking supervision can be observed.
 - At the international level, the **Basel Committee of Banking Supervision** (BCBS), as part of the Bank for International Settlement (BIS), is responsible for providing international regulatory guidelines.
 - These documents of the BCBS are not binding, but have to be translated into EU law, which is then binding or has to be incorporated into national legislation.
 - At the European level, banking supervision is carried out by the **European Central Bank** (ECB) in cooperation with national supervisors under the Single Supervisory Mechanism (SSM).
 - Based on the relevant European directives and regulations, the **European Banking Authority** (EBA) is developing a Single Rule Book.

- Time line of regulatory rules:
 - starting 1933 – Deposit protection schemes
 - 1988 – Basel I:
 - Uniform capital requirement for credit risk (8% rule)
 - 1989 – Rules for large exposures
 - 1990 – Rules for market risk
 - 1995 – Qualitative rules for the trading book
 - 1998 – New rules for market risk:
 - Internal models (Value-at-Risk (VaR) models) determining the capital requirements for the trading book.
 - 2002 – Qualitative rules for the banking book
 - 2004 – Basel II
 - Three pillar model (Risk sensitive minimum capital requirements / supervisory review process / market discipline)
 - 2005 – Minimum requirements for risk management
 - 2010 – Basel III
 - Capital definition and buffers / liquidity regulation / leverage ratio
 - 2014 – Bank recovery and resolution
 - 2017 – Basel IV (completing Basel III reform)

- Basel – Three Pillars



- Basel – Pillar 1: Minimum capital requirement
 - For important risks minimum capital requirements are defined, e.g.:
 - Credit risk
 - Market risk
 - Operational risk
 - Standardized approaches are used to (conservatively) estimate the risk.
 - Internal models can be used for this risk quantification given that certain quantitative and qualitative requirements are fulfilled.
 - There is a clear definition of the capital instruments that can be used to fulfill the capital requirements.
 - Legal reference: Regulation (EU) Nr. 575/2013 and 2019/876
 - Many guidelines exist defining qualitative requirements.

- Basel – Pillar 2: Supervisory Review Process
 - Principle-based regulation
 - Important elements for assessing risk:
 - Internal Capital Adequacy Assessment Process (ICAAP)
 - Internal Liquidity Adequacy Assessment Process (ILAAP)
 - The bank has to define its own methodology.
 - The bank has to define how to measure the available capital/liquidity.
 - The processes have to be adequate. There are no additional legal requirements.
 - ICAAP/ILAAP are the central process to manage the risks.
 - ICAAP and ILAAP are evaluated by the Supervisory Review and Evaluation Process (SREP).
 - This evaluation can lead to additional capital charges.
 - Legal reference: §§ 39 and 39a Austrian Banking Act (obligation to exercise diligence)
 - EZB Guideline „ICAAP“ is important document for implementation.

- Basel – Pillar 3: Market discipline:
 - Disclosure requirements:
 - Risk profile
 - Overview of risk management, key metrics and RWA
 - Comparison modelled and standardized RWA
 - Asset encumbrance
 - Capital distribution constraints
 - Should reflect ICAAP/ILAAP results.
- Legal reference: Regulation (EU) Nr. 575/2013 and 2019/876

Principles of bank-wide risk management (ICAAP)



Definition: Bank-Wide Risk Management

- Bank-wide risk management combines the assessment of the risk-bearing capacity of banks with risk/return considerations.
- This process covers the measurement and management of all material risks. Ensuring that the risk-bearing capacity is sufficiently high.
- Furthermore, the comparison of expected and realized profits with accepted risks is important.
- Bank-wide risk management is an important responsibility of the management.
- With the implementation of Basel II, this concept became an important part of the regulatory framework for banks.

Definition: Bank-Wide Risk Management

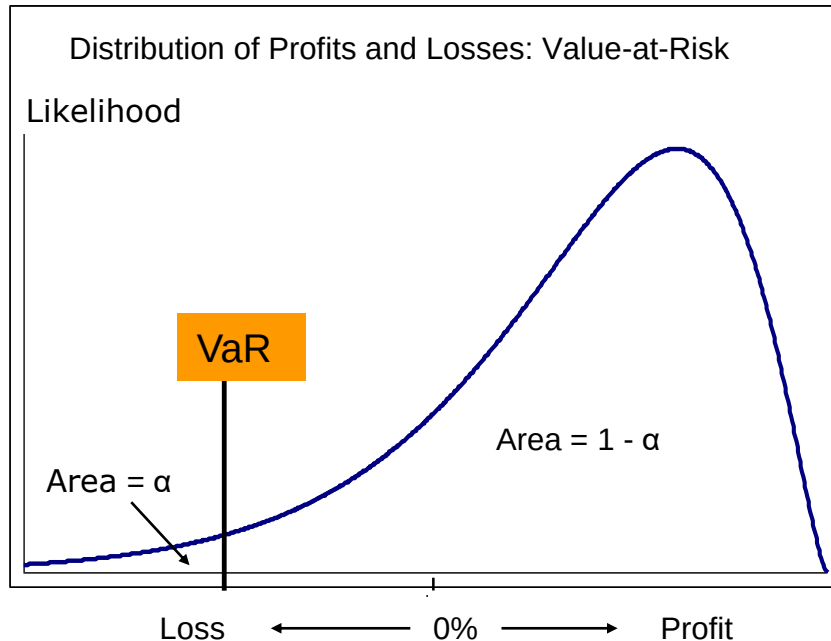
- Bank-wide risk management has changed over time.
- Due to the possibility of trading more risk types (e.g. introduction of credit derivatives), the risk position of banks can be quickly changed/adjusted.
- Therefore a timely management of risks and profits is necessary.
- This is a challenging task concerning the availability of data and the IT infrastructure.
- A modern bank needs comprehensive data warehouses and efficient IT systems to offer a management cockpit that allows a clear analysis of risks and profits.

Basic Concept of Risk Management

Assets	Liabilities
Cash	Deposits
Loans	Bonds
Other Assets	Equity

- The overall risk of the assets and liabilities has to be estimated.
- This risk measure can be compared with the expected return -> allows to manage business activities.
- On the other hand, this risk measure can be compared with the available equity -> allows to manage the bank's default probability.

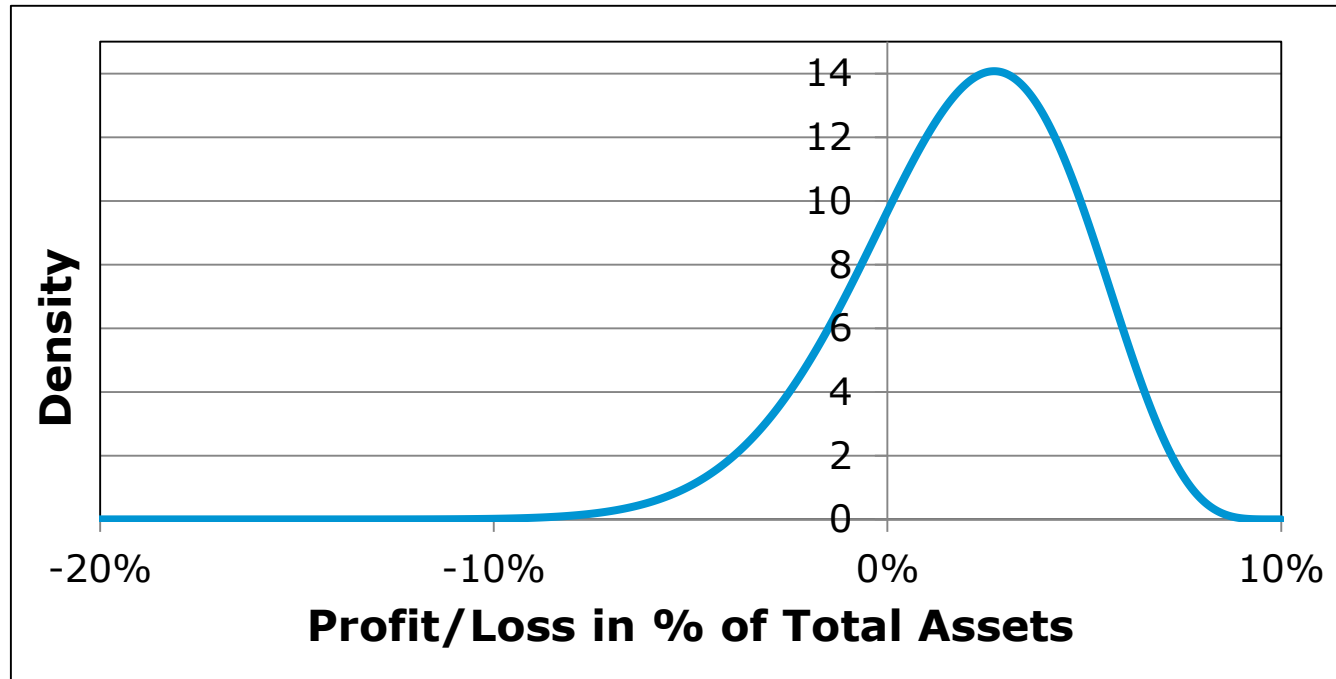
Basic Concept of Risk Management



- Value-at-Risk (VaR) is the loss, which will only be exceeded with a probability of α .
- VaR is a risk measure, which is often used for risk management purposes, e.g., in important regulatory models.
- Alternative measures provide additional risk-specific information, e.g., expected shortfall.

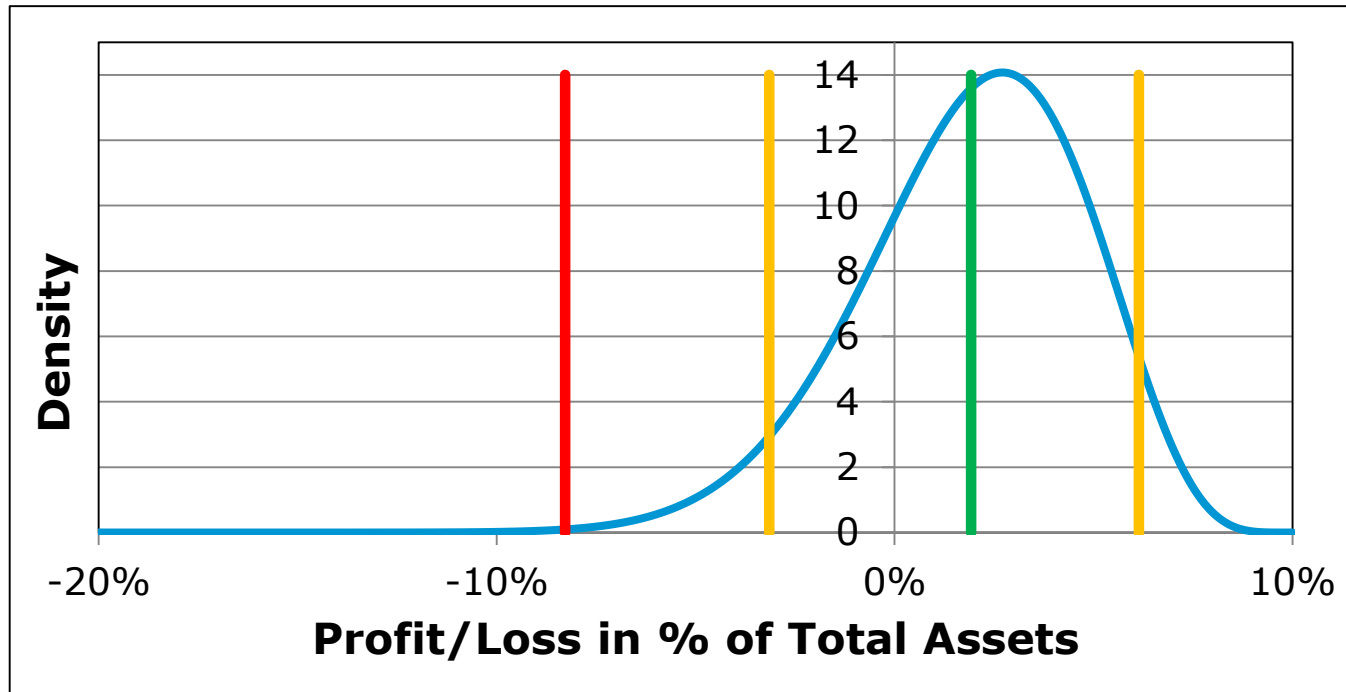
Basic Concept of Risk Management

- Starting point is the risk measurement: In the best case the overall risk position of a bank over a certain time period can be represented as a probability distribution of profits and losses.



Basic Concept of Risk Management

- Key figures can be determined based on this distribution. These figures are important in the bank-wide risk management.



Basic Concept of Risk Management

- The following figures allow to represent the risk position:
 - Expected Profit/Loss: 1.92%
 - 90% confidence interval: -3.15%; 6.14%
 - 99.9% Value-at-Risk (i.e., the 0.1% quantile): -8.28%
- The risks can be managed based on these figures:
 - Expected profits can be compared with risks. The pricing of financial contracts becomes possible.
 - The risk position can be compared to the available internal capital. This comparison allows to determine the risk-bearing capacity.
 - The figures allow to define limits for the risk position.
 - The risk measurement can be carried out for different areas. This allows to compare different units or products with each other.

Basic Concept of Bank-Wide Risk Management

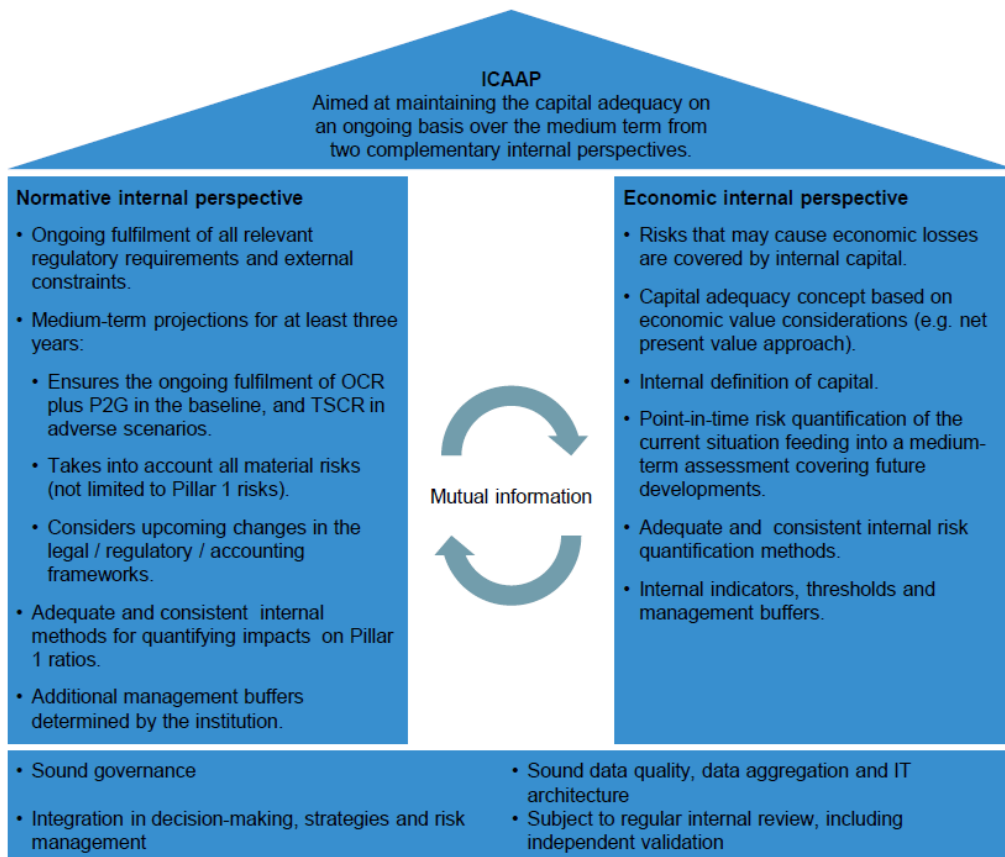
- The pillar 2 of Basel requires that banks introduce an ICAAP (internal capital adequacy assessment process) to ensure that banks can evaluate their risk-bearing capacity.
- In the Austrian banking law this requirement is formulated as part of the obligation to exercise diligence for the management.
- The models and processes can be chosen by the bank. The only formal requirement is the appropriateness of these methods.
- Two perspectives have to be considered:
 - Economic perspective
 - Normative perspective

Basic Concept of Bank-Wide Risk Management

- Economic perspective:
 - The goal is to identify and quantify all material risks.
 - The quantification is focused on economic value considerations (i.e., estimation of economic losses, not balance-sheet losses).
 - These risks should be covered with internal capital (economic capital adequacy).
 - One possibility is to use the 99,9% VaR for a one year period.

- Normative perspective:
 - Again, all material risks should be considered.
 - Institution's ability to fulfil all of its capital-related quantitative regulatory and supervisory requirements should be analyzed (e.g. minimum capital requirements).
 - A multi-year period should be covered (minimum of three years).
 - A baseline and multiple adverse scenarios should be covered.
 - The risk quantification methods can be different compared to economic perspective and not every scenario has to cover adverse effects in all risk types.

Basic Concept of Bank-Wide Risk Management

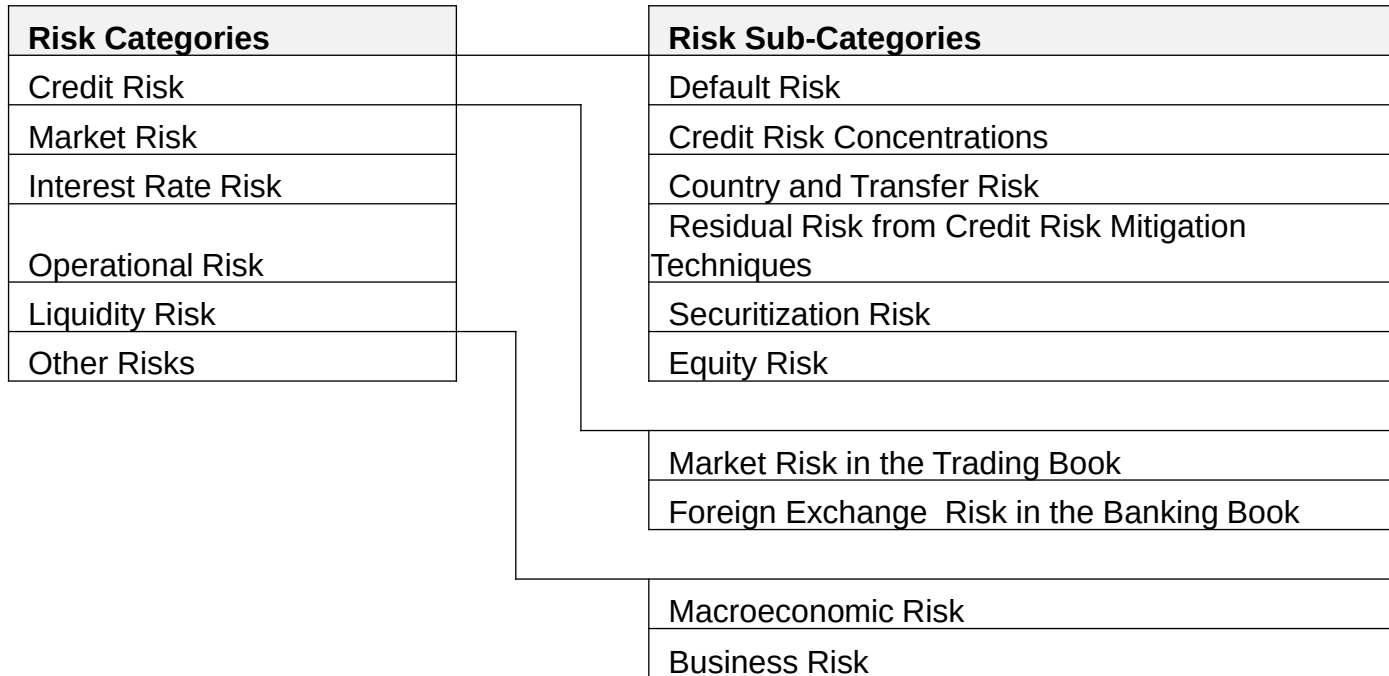


Basic Concept of Bank-Wide Risk Management

- Although, bank-wide risk management in general is a central task of the management, the introduction of the pillar 2 of Basel II has lead to a modernization of this process.
- Regulatory guidelines are available on how to implement a bank-wide risk management (Internal Capital Adequacy Assessment Process – ICAAP). These guidelines offer a good overview for a state-of-the-art bank-wide risk management.
- Every bank needs to apply the following steps within the bank-wide risk management process:
 - Risk identification
 - Risk quantification
 - Risk aggregation
 - Capital allocation
 - Risk monitoring (risk-bearing capacity, limits, stress tests)
 - Risk management (pricing, performance measurement, strategic decisions)

- The risk identification is an important prerequisite for the bank-wide risk management.
- In this step a classification of risk categories is necessary. Based on this classification an assessment of risk categories has to be carried out.
- This is a central step for the introduction of the ICAAP, as for this setup a first, simple quantification of risks is necessary to assess which risk categories are material.
- This step also reveals on which information/data the bank can base such assessments.
- The risk identification can also trigger a discussion in the bank, which risks the bank wants to take.

Risk Identification: Risk Categories



Risk Identification: Information Sources

- The internal and external reports of the bank form a good starting point for the risk identification.
- Furthermore, mandatory reports for the regulator contain important information.
- In most cases the analysis of these reports allow to identify the most important risks that are material.
- The result of this analysis should be discussed with all relevant departments and the management of the bank to finalize the assessment.
- The material risks should be further categorized into low / medium / high significance.

Risk Identification: Example

Risk Category	Risk Indicator Level
Default Risk	medium
Credit Risk Concentrations	low
Country and Transfer Risk	not material
Residual Risk from Credit Risk Mitigation Techniques	not material
Securitization Risk	not material
Equity Risk	low
Market Risk in the Trading Book	high
Foreign Exchange Risk in the Banking Book	high
Macroeconomic Risk	medium
Business Risk	high

Risk Identification: Case Study

- Reports for regulatory purposes form the starting point.
- Analyze the provided information and answer the following questions:
 - Which risk categories are material?
 - Which risk indicator levels would you choose for the material risks?
 - Is there information missing?
 - With which departments would you discuss these results?

Risk Identification: Important Issues

- The risk identification has to use more information than simply the minimum capital requirements.
- A detailed analysis of the risk categories is necessary.
- Starting point is the actual risk structure additional considerations are necessary:
 - What risks can the bank take anytime based on which limits? (If there is no open FX risk position, this does not imply that no department has the limit to open such a position.)
 - How can be assured that non-material risks stay non-material? Are there limits in place?
 - Comparison of actual and target risk structure
 - Discussion of results with management

- For each risk category that is considered material after the risk identification, a methodology for quantifying this risk has to be chosen.
- Important for this step is the principle of proportionality.
- There are two aspects of proportionality:
 - Size of the bank
 - Importance of the risk factor
- The smaller the bank and the less important the risk factor is the simpler can be the risk quantification methodology.
- In the simplest case the method to estimate the minimum regulatory capital can be used. In the most complex case a stochastic model (based on simulation techniques) has to be applied.
- The criteria for the selection of the method has to be clear.

- Risk quantification implies to have models and processes in place to estimate the distribution of the risk factor and relate this distribution to profits and losses for the bank.
- Risk is defined as the potential deviation from the expected outcome resulting in lower profits or losses.
- In the most cases the risk is evaluated by estimating the 0.1% quantile of the distribution for a one year time period (Value-at-Risk calculation.)
- In the simplest case this loss defines the minimum capital level that has to be available to cover this risk.
- Simple methods can rely on scenario analysis, i.e. a certain scenario is assumed to represent the 0.1% quantile.

Risk Quantification: Credit Risk

- Credit risk refers to the negative consequences associated with defaults or the non-fulfillment of concluded contracts in lending operations due to a deterioration in the counterparty's credit quality.
- As lending is considered the main operation of banks, this risk category will be material for most banks.
- Depending on the business model credit risk can be low (diversified portfolio containing high quality borrowers) to high (undiversified portfolio of risky borrowers).
- When quantifying credit risks, in most cases, two aspects will be important:
 - Default risk
 - Credit risk concentrations
- These two aspects also allow to discuss the principle of proportionality.
- This has to be carefully considered as the modeling of credit risks is only possible with an extensive access to data and sophisticated models.

- The simplest method for evaluating credit risks is the Standardized Approach of Basel II:
 - The capital requirement of a loan is based on the external rating.
 - When no external rating is available the capital requirement is 8%.
- In general, it will be necessary to estimate the following risk parameters for each obligor to quantify the credit risk:
 - Probability of Default (*PD*)
 - Loss Given Default (*LGD*)
 - Exposure at Default (*EAD*)

- Probability of default (PD):
 - Default intensity (actuarial likelihood).
 - Asset value, NPV of household income, public debt / deficit,....

- Loss given default (LGD):
 - Expected recovery quota after bankruptcy.
 - Collateral, netting opportunities, guarantees,....

- Exposure at Default (EAD):
 - Current exposure.
 - Open lines, overdraft facilities, contingent exposures (eg, letter of credit),...

- These risk parameters can be estimated based on
 - Historic default data
 - Market values based on pricing models
- Pricing models either
 - Model default as an exogenous event, e.g. reduced-form models.
 - Model default based on underlying factors, e.g. obligor defaults if asset value falls below a certain barrier – Merton model.
- To estimate the profit/loss distribution correlation effects have to be considered:
 - E.g. it could be assumed that the asset values of two obligors are correlated
- This idea is used in the IRB Advanced approach of Basel II:
 - This approach uses PD, LGD and EAD as input and adds a correlation structure by assuming correlated asset values.

- The IRB approach is risk sensitive, however, the following assumptions are used:
 - simple correlation structure is assumed
 - completely diversified portfolio is assumed
- If these assumptions are not fulfilled, then more complex credit risk models have to be used, especially, if concentration risks are an issue.
- The development of such models is complex, i.e. simulation models based on the available risk parameters can be used.
- Most banks will at least need credit risk models similar to the IRB approach.
- For small banks this is a difficult task.

Risk Quantification: Market Risk

- In most cases three areas are important when quantifying market risk:
 - Market price risk in the trading book
 - FX risk in the banking book
 - Interest rate risk in the banking book
- The choice of the models will depend on the risk profile of the bank:
 - Standard approaches / Interest rate statistic
 - Scenario analysis
 - Value-at-Risk methods
- It will depend on the trading activity which approach is appropriate.
- It is very important to assure that risk positions that cannot be handled by the approach are not possible.

- Example Interest Rate Risk in the Banking Book:
 - The standard approach to evaluate interest rate risk is to calculate the loss in market price for parallel shifts of ± 200 basis points in the interest rate curve.
 - For a simple plain-vanilla bond portfolio this will likely be sufficient.
 - If interest rate swaps and interest rate options can be traded then (as an extreme example) a position can be build that has a loss of zero for the parallel shift of the interest curve.
 - In most cases scenarios for twists of the interest rate curve can be added to quantify such risks. These were added to the regulatory requirements.
 - For complex portfolios (e.g. positions on changes in the interest rate volatility) VaR models are necessary.

Risk Quantification: Operational Risk

- Operational risk is defined as the risk of loss resulting from inadequate or failed internal processes, people or systems, or from external events.
- Different approaches are available to measure operational risk. The simplest approach is the Basic Indicator Approach. In this approach, a risk weight of 15% is applied to a single indicator, specifically the average gross income (i.e. the sum of net interest income and net non-interest income) over the last three years.
- More advanced approaches rely on the availability of a loss database.
- Even if a loss database is not available, it is important to have data on the largest losses due to operational risk available.

Risk Quantification: Liquidity Risk

- Liquidity risks can be categorized as term liquidity risk, withdrawal/call risk, structural liquidity risk (funding liquidity risk) and market liquidity risk.
- Banks can assess the significance of liquidity risks by comparing liquid or easily liquidated balance-sheet assets with short-term liabilities.
- In principle, it is not mandatory to hold capital against liquidity risk.
- It is important to have processes in place ensuring the liquidity of the bank (ILAAP).
- Basically, a detailed analysis of the structure of short-term assets and liabilities is necessary (e.g. analysis of pre-payment-risk).
- Furthermore, emergency plans should be available (during the financial crisis most banks experienced liquidity shortage.)

Risk Quantification: Other Risk

- The quantification of the other risks is very difficult. The following risk categories banks should consider:
 - Macroeconomic Risk
 - Business Risk
- Due to the lack of available quantitative measurement methods, other risks can also be measured and controlled by exclusively qualitative means.
- In order to incorporate these risks into its risk management, the bank can maintain a capital cushion for other risks.

- Overall bank risk is a combination of all the risk types which are relevant to the individual bank.
- The overall risk position of the bank has to be calculated by means of aggregation.
- For banks which rely heavily on the basic methods of calculating minimum capital requirements in their assessment of individual risk types, aggregation can be performed by simply adding the risks calculated.
- This is a conservative form of aggregation as it is implicitly assumed that risks are perfectly correlated.
- Using more sophisticated models of aggregation allows to consider diversification effects.
- Especially, the aggregation of market and credit risk is an interesting problem.

- In this perspective, the risk of a negative change in the economic value is determined.
- This change is compared with the bank's economic capital.
- Thus, a risk-bearing capacity analysis is performed to ensure that the full economic capital is only lost with a low probability.
- The 99.9% VaR for a one-year horizon can be considered the benchmark value here.
- Depending on their business model, banks can use stricter risk measures.
- There are different definitions of economic capital, often the net asset value is used.

Economic Perspective: Risk Quantification

Risk Category	Risk Indicator Level	Pillar 1	Pillar 2: VaR 99,9%
Default Risk	medium	80.000.000	80.000.000
Credit Risk Concentrations	low		25.000.000
Country and Transfer Risk	not material		
Residual Risk from CRM Techniques	not material		
Securitization Risk	not material		
Equity Risk	low		5.000.000

Market Risk	high	20.000.000	30.000.000
FX Risk	high		20.000.000
Credit Spread Risk	not material		

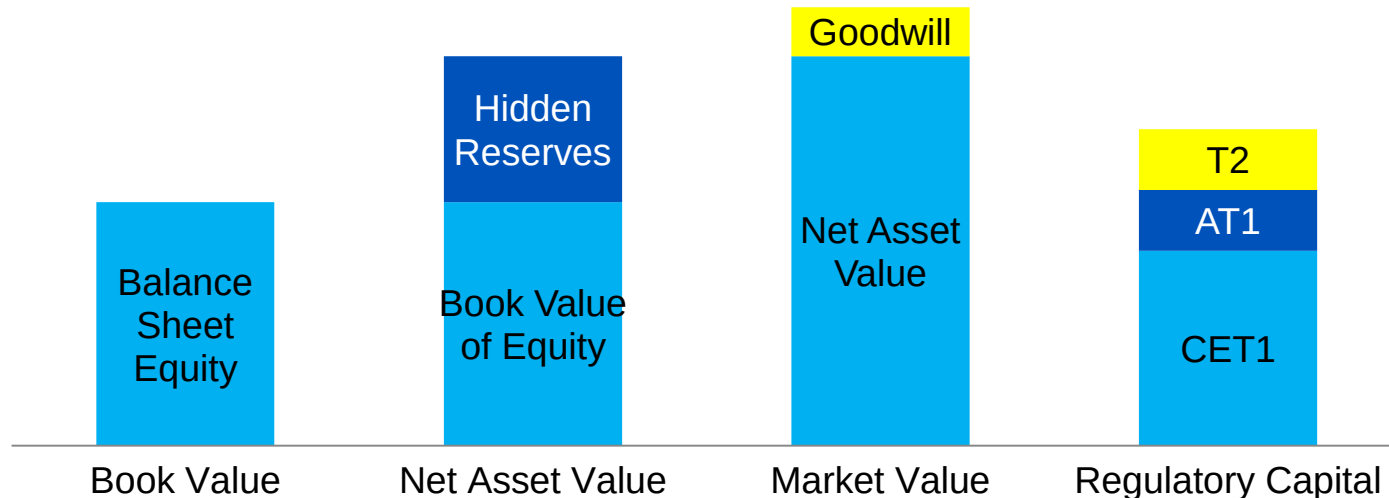
Interest Rate Risk	medium		8.000.000
Operational Risk	medium	15.000.000	15.000.000
Liquidity Risk	low		3.000.000

Macroeconomic Risk	medium		5.000.000
Business Risks	high		5.000.000

Economic Perspective: Definition of Internal Capital

- Different types of internal capital are available to cover potential losses:

Categorization of Equity Capital



Economic Perspective: Risk-bearing capacity

- The analysis of the risk-bearing capacity consists in the comparison of the aggregate risks with the internal capital.
- As long as the overall risk position of the bank is lower than the available economic capital the bank can bear its risks.
- This comparison provides also the possibility to set a limit on the overall risk level.
- This limit has to be monitored and a breach of the limit should result in risk reducing actions.
- A traffic light approach can be used.
- The limiting of risk should also be done for individual business units and risk types.
- Management buffers can be used to allow for more flexibility.

Economic Perspective: Risk-bearing capacity

Risk Category	Risk Level Indicator	Pillar 1	Pillar 2: VaR 99,9%
Default Risk	medium	80.000.000	80.000.000
Credit Risk Concentrations	low		25.000.000
Equity Risk	low		5.000.000

Market Risk	high	20.000.000	30.000.000
FX Risk	high		20.000.000

Interest Rate Risk	medium		8.000.000
Operational Risk	medium	15.000.000	15.000.000
Liquidity Risk	low		3.000.000

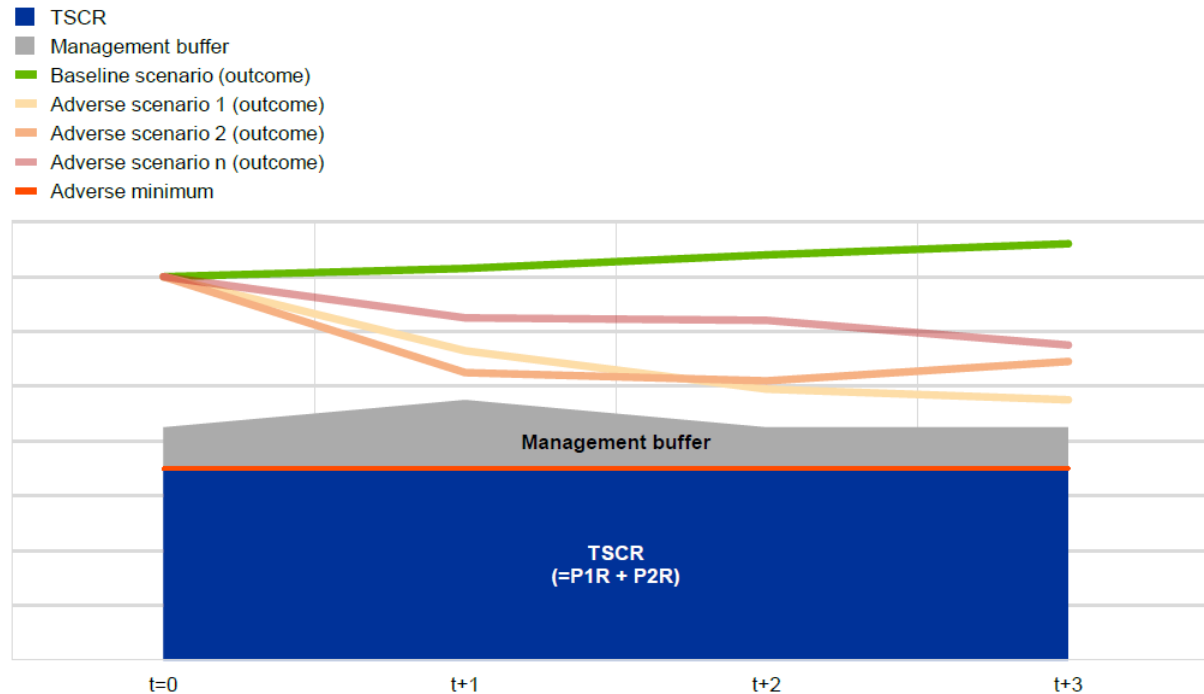
Macroeconomic Risk	medium		5.000.000
Business Risk	high		5.000.000
Aggregated Risk		115.000.000	196.000.000
Limit		130.000.000	210.000.000
Economic Capital		160.000.000	220.000.000

- In the normative perspective, it is analyzed whether quantitative regulatory and supervisory requirements can be fulfilled in adverse scenarios in a multi-year time period.
- The scenarios considered should reflect the business model and the specific setup of the bank.
- Not every scenario must consider losses in all material risk categories.
- Some scenarios can also rely on simple analyses. The probability of occurrence does not always have to be exactly measurable.
- However, the relevance of these scenarios should be clear.
- The risk measurement methods of the economic perspective can also be included.

- The scenarios analyze how the fulfilment of regulatory requirements changes.
- There are multiple requirements that should be considered:
 - Minimum capital requirements
 - Leverage ratio
 - Liquidity ratios
- The focus is often on the fulfilment of the minimum capital requirements.
- Here the banks consider the total SREP capital requirement (TSCR).
- The time horizon is often three years
- Management buffers can also be considered in this perspective.

Normative Perspective

- Example in the ECB guide:



- An adverse scenario, which includes all types of risk, can be determined with the existing risk measurement in the economic perspective.
- Here, a loss can be analyzed that is realized in the first year and its effects are then analyzed over the next years.
- For this purpose, a going-concern loss quantile can be defined (e.g. 95% VaR) and it can be assumed that the total loss is also reflected in the regulatory ratios.
- Since this is not always the case, there are usually numerous adjustments.
- In determining whether, for example, the TSCR is complied with, realistic compensatory measures can still be taken into account in principle, e.g. hidden reserves that can be realized short term.

Normative Perspective

Risk Categories	Risk Level Indicator	Pillar 2: VaR 95%
Default Risk	medium	25.000.000
Credit Concentration Risk	low	5.000.000
Equity Risk	low	1.000.000
Market Risk	high	5.000.000
FX Risk	high	3.000.000
Interest Rate Risk	medium	2.000.000
Operational Risk	medium	4.000.000
Liquidity Risk	low	1.000.000
Macroeconomic Risk	medium	1.000.000
Business Risk	high	1.000.000
Aggregated Risk		48.000.000
Internal Capital		220.000.000
Internal Capital after Losses		172.000.000
Limit		165.000.000
TSCR		160.000.000

Risk Monitoring: Limits

- The comparison of the risk and capital allows to define limits for the risk position of the bank.
- A limit for the aggregated risks can be defined at the bank level.
- This limit has to be monitored and if the limit is exceeded the risk level has to be reduced.
- Traffic-light-approaches can be used to define green, yellow and red levels of limit utilization, i.e. defining early warning levels to react in time.
- It is important to limit the risks not only at the overall level of the bank, but define individual limits for risk categories or business units.
- These limits have to be in line with the overall limit for the aggregated risks.

- The introduction of a limit system is the central prerequisite for risk monitoring (no risk without a limit).
- The individual limits of the risk categories have to be chosen in a way that with a 100% utilization of the individual limits the overall bank limit is not exceeded.
- For some units structural limits should be used in addition to capital limits.
- Structural limits are defined based on characteristics of financial instruments, e.g. volume, rating, maturity
- Again the compliance with the structural limits should guarantee the compliance with the capital limits.
- The advantage of structural limits is that these limits can be communicated more efficiently within the bank.

Risk Monitoring: Stress Tests

- In addition to the risk monitoring based on the presented methodologies banks have to have stress tests in place to analyze the risk-bearing capacity.
- Stress tests show the effects of events which can not (or not sufficiently) be accounted for in “normal” risk measurements (e.g. with VaR methods).
- These exceptional scenarios should be analyze based on stress tests.
- These scenarios can be based on historic events or defined by experts within the bank.
- The objective of the stress test is not that the bank should survive any extreme event, but it should trigger a discussion process within the bank.

- The results of the ICAAP should be used in the development of the business strategy of the bank and in the management decisions.
- In a first step the actual risk structure can be compared to a target risk structure.
- This can trigger a focused discussion about the risk position of the bank.
- The first step to manage the risk can be the definition or refinement of the limit system.
- However, a significant benefit for the bank can only be achieved, if the results of the ICAAP is used for actively managing the business including a comparison of profits and risks.
- Basically the results of the ICAAP can be used to identify the business units that achieve the highest risk-adjusted profits. These business units can be potentially extended.

Risk Management: Risk-adjusted Performance

- Different measures can be used to compare risk and return:
 - RORAC (Return on Risk-adjusted Capital)
 - RAROC (Risk-adjusted Return on Capital)
 - EVA (Economic Value Added)
- Example RORAC:
- Without risk-adjustment only the ROI (return on investment) can be used

$$\text{RORAC} = \frac{\text{Return}}{\text{Risk Capital}}$$

$$\text{ROI} = \frac{\text{Return}}{\text{Invested Volume}}$$

Risk Management: Loan Pricing

- The price of products has to be defined in a way that assures that all relevant costs are considered and all risk-related components are priced.
- For example: A typical loan pricing has the following components:
 - Refinancing costs, e.g. 3M Euribor rate
 - Operating costs
 - Expected Loss
 - Risk premium (capital costs):
 - Required capital · cost of equity
- The expected loss and risk premium are based on the estimated credit risk parameters and assure that in an average year shareholders earn the cost of equity.

Credit Risk Measurement and Validation



Introduction – Credit Risk Measurement

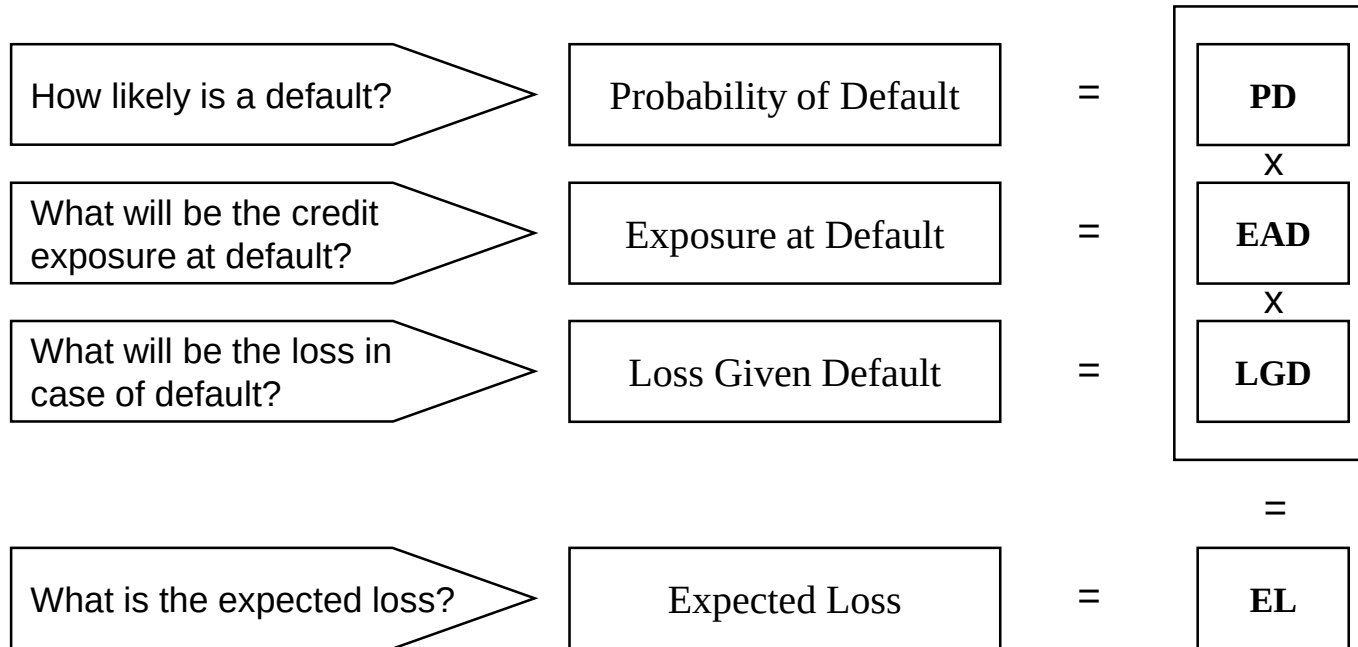
- The risk estimation in a typical bank is not performed at the overall bank level, but individually for the specific risk types.
- Thus, typically specific models are in place to estimate the credit risk of the bank.
- Thus, a probability distribution for credit risk is estimated.
- These estimations consists of various building blocks:
 - Risk parameter for the individual borrowers
 - Correlation between borrowers
 - Credit portfolio model

Introduction – Credit Risk Measurement

- Credit risk is the risk of loss due to a obligor's non-payment of a loan or other line of credit (either the principal or interest (coupon) or both).
- Default: debtor has not met its legal obligations according to the debt contract, e.g. it has not made a scheduled payment.
- Credit risk measurement is the prerequisite for
 - Granting and pricing of credit risk products (e.g. loans)
 - Management of credit portfolio
 - Risk management
 - Banking regulation (calculation of regulatory capital requirements)

Introduction – Credit Risk Measurement

- Dimensions credit risk:



Introduction – Credit Risk Measurement

- Example: A retail loan has an exposure of 20,000 EUR and has to be repaid in on year. The probability of a default is 1% for this period and the loss is 50% of the exposure.

Estimate the expected loss of the loan.

Provide the probability distribution.

Split the exposure in two separate loans. Repeat the exercise

Probability of Default (PD)

Introduction – Probability of Default

- Probability of Default (PD): The Probability of Default is the likelihood that a loan will not be repaid and fall into default within a certain period of time.
- How to define default?
 - Economic basis:
 - Missed payment
 - Restructuring with loss (in present value)
 - High probability for financial distress
 - Legal basis:
 - Provisions
 - Filing for bankruptcy
 - Similar protection (Chapter 11)

Please note: The default definition has to be identical for all risk parameters (PD and LGD).

Introduction – Probability of Default

- The Basel regulation provides the following definition of default (CRR: Regulation (EU) 575/2013 Article 178)

A default shall be considered to have occurred with regard to a particular obligor when either or both of the following events has taken place:

(a) the credit institution considers that the obligor is unlikely to pay its credit obligations to the credit institution, the parent undertaking or any of its subsidiaries in full without recourse by the credit institution to actions such as realizing security (if held);

(b) the obligor is past due more than 90 days on any material credit obligation to the credit institution, the parent undertaking or any of its subsidiaries.

Introduction – Probability of Default

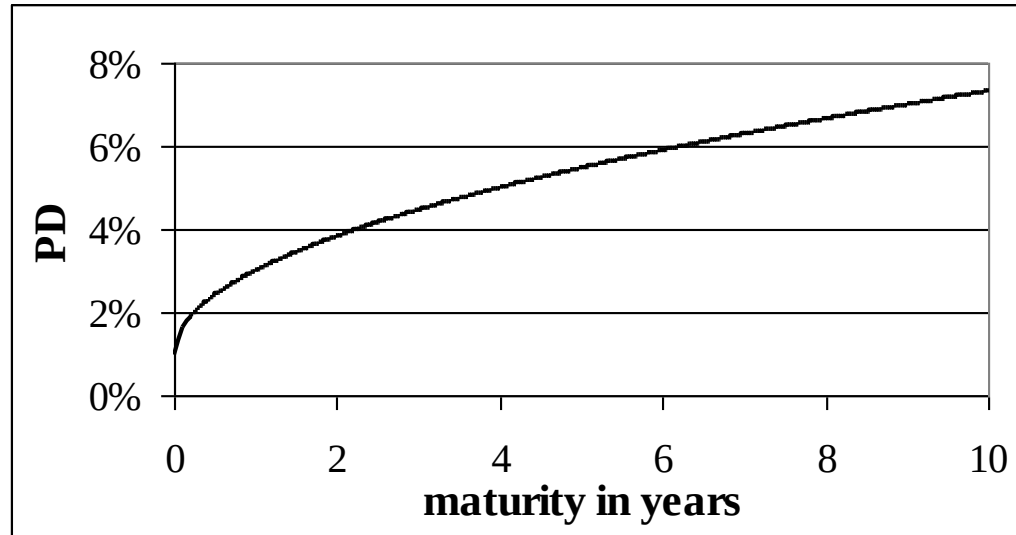
- How shall the “period of time” for the PD estimation be defined?
 - A credit institutions has bought a corporate bond with a two year maturity.
 - The credit institutions plans to sell the bond in one year.

Question: Is the knowledge of the one year default probability sufficient to assess the risk of the position?

Answer: No. If the corporations does not default within the first year, the price of the bond after one year does depend on the default probability within the second year.

Introduction – Probability of Default

- In the ideal case the default probability is available for each maturity (term structure of PD):



- Most credit institutions base their PD estimation on ratings. These ratings are often calibrated on one year default rates.
- Credit institutions can either develop an internal rating process or rely on an external rating process.
- Internal rating process: Own rating methodology for classifying the obligors into rating classes has to be developed. Based on rating and default history PDs can be estimated.
- The CRR contains various rules / requirements that have to be fulfilled by internal rating models.
- External rating process: Alternatively banks can rely on rating agencies (Standard&Poor's, Moody's, Fitch,...). These agencies provide the classification of the customers into rating classes and the PDs for these classes.

Rating Process

- Obligors are classified into the following ratings by S&P / Moody's:

AAA	Aaa	Highest rating class. The obligor's capacity to meet its financial commitment on the obligation is extremely strong.
AA	Aa	The obligor's capacity to meet its financial commitment on the obligation is very strong.
A	A	An obligation rated 'A' is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than obligations in higher-rated categories. However, the obligor's capacity to meet its financial commitment on the obligation is still strong.
BBB	Baa	An obligation rated 'BBB' exhibits adequate protection parameters. However, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity of the obligor to meet its financial commitment on the obligation.
BB	Ba	An obligation rated 'BB' faces major ongoing uncertainties or exposure to adverse business, financial, or economic conditions which could lead to the obligor's inadequate capacity to meet its financial commitment on the obligation.
B	B	An obligation rated 'B' is more vulnerable to nonpayment than obligations rated 'BB', but the obligor currently has the capacity to meet its financial commitment on the obligation. Adverse business, financial, or economic conditions will likely impair the obligor's capacity or willingness to meet its financial commitment on the obligation.
CCC	Caa	An obligation rated 'CCC' is currently vulnerable to nonpayment, and is dependent upon favorable business, financial, and economic conditions for the obligor to meet its financial commitment on the obligation. In the event of adverse business, financial, or economic conditions, the obligor is not likely to have the capacity to meet its financial commitment on the obligation.
CC	Ca	An obligation rated 'CC' is currently highly vulnerable to nonpayment.
C	C	A subordinated debt or preferred stock obligation rated 'C' is currently highly vulnerable to nonpayment. The 'C' rating may be used to cover a situation where a bankruptcy petition has been filed or similar action taken, but payments on this obligation are being continued.
D	D	Default

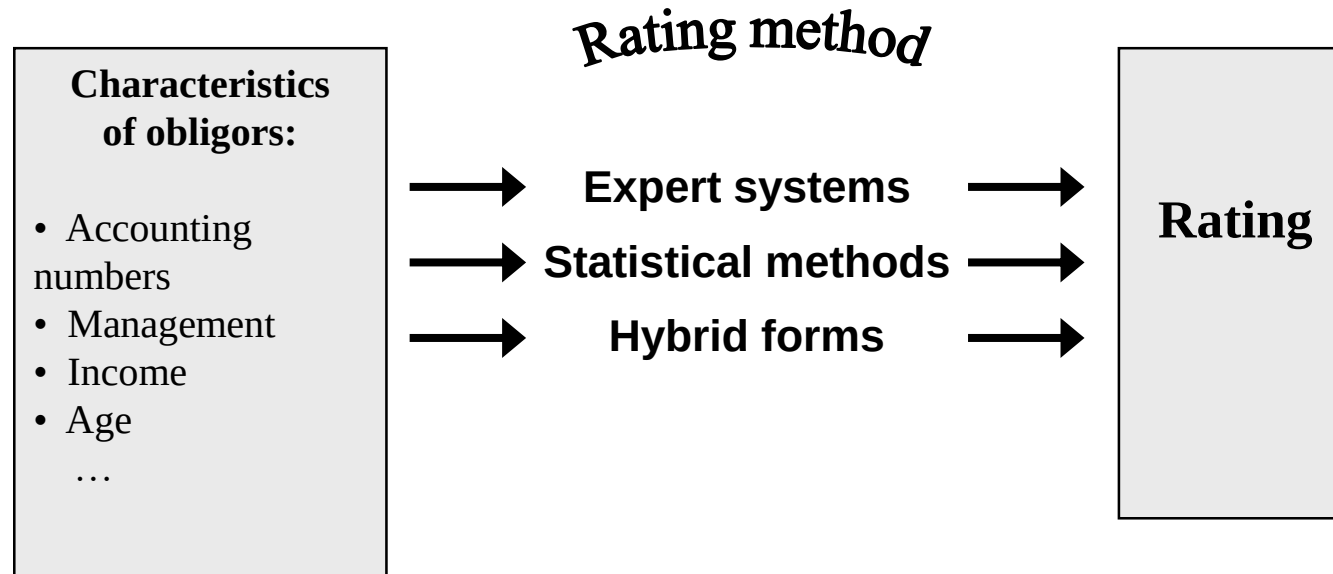
Rating Process

- One year transition matrix of Standard&Poor's:

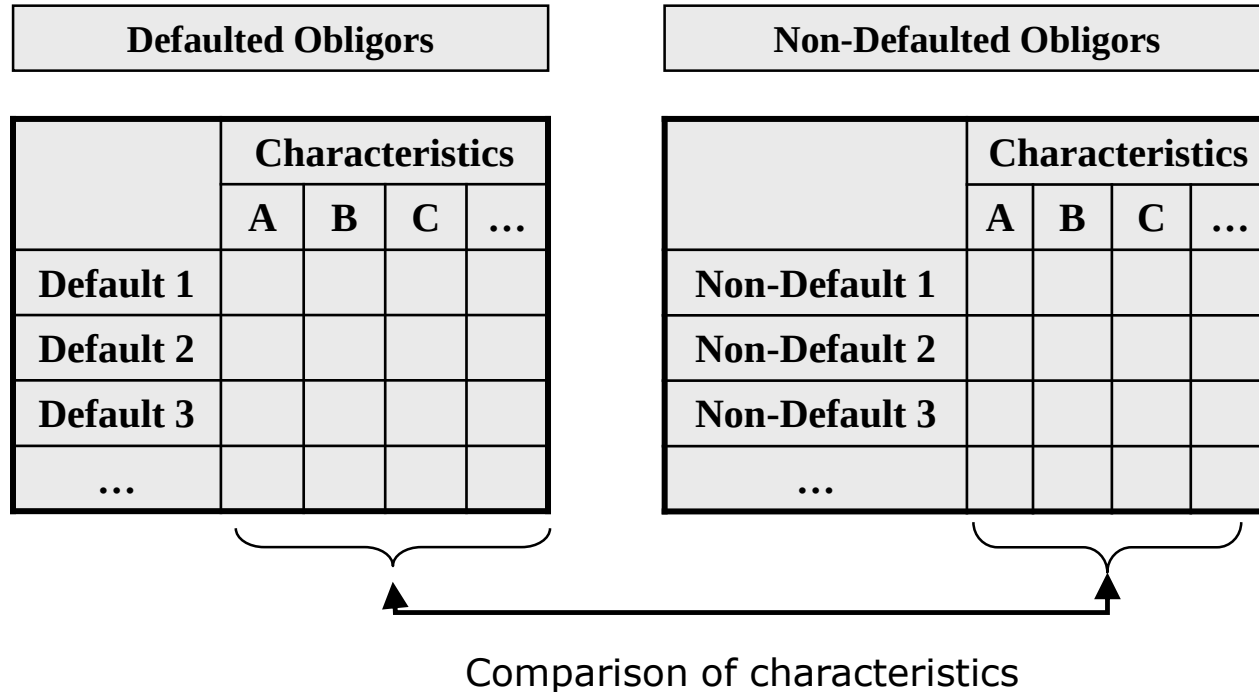
From / To	AAA	AA	A	BBB	BB	B	CCC/C	Default
AAA	92,08%	7,09%	0,63%	0,14%	0,06%	0,00%	0,00%	0,00%
AA	0,62%	90,83%	7,77%	0,59%	0,06%	0,10%	0,02%	0,01%
A	0,05%	2,09%	91,38%	5,79%	0,44%	0,16%	0,04%	0,05%
BBB	0,03%	0,21%	4,09%	89,38%	4,82%	0,86%	0,24%	0,37%
BB	0,03%	0,08%	0,40%	5,53%	83,25%	8,15%	1,11%	1,45%
B	0,00%	0,08%	0,27%	0,34%	5,39%	82,41%	4,92%	6,59%
CCC/C	0,10%	0,00%	0,29%	0,58%	1,55%	10,54%	52,80%	34,14%
Default	0,00%	0,00%	0,00%	0,00%	0,00%	0,00%	0,00%	100,00%

- Advantage of external rating process:
 - Information is easy to access for credit institution
 - No data collection necessary
 - Default probabilities are estimated based on a high number of obligors
 - Ratings may be more objective
- Disadvantage of external rating process:
 - Only a small number of potential obligors has an external rating
 - PDs in the portfolio of the credit institution may be different
 - Number of rating classes is given
 - Default definition is given
 - Maturity and risk parameter (PD or EL) not communicated

- For these reasons most credit institutions have developed internal rating processes for most customer groups:



- Rating methods are developed based on the comparison of defaulted and non defaulted obligors:



- The development of a rating method (rating model) consists of several steps:
 - Definition of characteristics that potentially influence the probability of default
 - Hypotheses of the effect of characteristics (e.g. higher ROE → lower PD)
 - Data collection: characteristics and subsequent defaults
 - Univariate analysis whether the characteristics have significant effect on PD
 - Univariate analysis whether stated hypotheses are correct
 - Specification of multivariate model
 - Use of model output (e.g. credit score) for classification of obligors into the defined rating classes

- Regressions:
 - For obligor i default/non-default can be modeled using the following random variable:

$$Y_i = \begin{cases} 1 & \text{obligor } i \text{ defaults} \\ 0 & \text{obligor } i \text{ does not default} \end{cases}$$

- The probability distribution of this bivariate random variable has the following form:
 - Probability for state 1 (default): PD_i
 - Probability for state 0 (no default): $1 - PD_i$
- The expectation for Y_i is then given as:

$$\text{Expectation}[Y_i] = 0 \cdot (1 - PD_i) + 1 \cdot PD_i = PD_i$$

- Logistic Regression:
 - State-of-the-art rating method for practical applications
 - The logistic regression (Logit-Model) is similar to the linear regression, but avoids the disadvantages of the linear regression:
 - PD is guaranteed to be in the interval [0%, 100%]
 - Influence of the characteristics is not modeled to be linear
 - The logistic regressions uses the same data structure as the linear regression.
 - Since the characteristics have no longer a linear influence, the parameter estimation is more complex.

- The logistic model is based on the logistic distribution function as link function:

$$G(x) = \frac{e^x}{1 + e^x} = \frac{1}{1 + e^{-x}}$$

- The probability of default is therefore defined as the following function based on the characteristics:

$$PD_i = \frac{1}{1 + e^{-(\alpha + \beta_1 \cdot X_{i1} + \dots + \beta_k \cdot X_{ik})}}$$

- The parameters are estimated by applying a maximum likelihood estimation.

- Based on the result of the logistic regression the obligor can be classified into one of the defined rating classes:
 - Each rating class is defined by a PD-interval, e.g.:
 - Rating class 1: PD between 0% and 0.5%
 - Rating class 2: PD between 0.5% and 2%
 - ...
 - Rating class u: PD between 30% and 100%
- Basically, the PDs could also be directly used, but in most cases transition matrix methods are applied, because time series information can be easily integrated and the development sample of the regression might have a non-representative absolute PD.

- Credit institutions started with logistic regression approaches, because the effect of characteristics on the probability of default is not a black box.
- It can be expected that in the next years more advanced methods will be applied to default data, e.g.
 - Neuronal networks
 - Support vector machines
 - Random forest
 - ...

Rating – Expert Systems

- For some obligor groups (e.g. specialized lending) not enough data will be available to calibrate statistical methods.
- For these obligors the ratings can be based on an expert judgment.
- The classification in rating classes can be either in a direct way or via a rating sheet:

Rating Sheet for Specialized Lending SPV (Assess the following characteristics between 1 and 10)	
Characteristic	Score
Profitability	3
Growth potential	4
Quality of management	2
...	...

- For many obligor groups (e.g. corporations) hybrid ratings representing a combination of statistical methods and expert systems can be found.
- In most cases the first (preliminary) result is based on statistical methods (representing the “hard facts”).
- In the second step an expert can modify this result based on her experience according to defined rules (e.g. one rating class up or down; up to two rating classes down, ...).
- Again the decision of the expert can be based on a verbal explanation or a rating sheet.

Probability of Default based on Ratings

- In general, credit institutions have a rating process for the obligors in place.
- The rating process classifies the obligor in one of the defined rating classes ("the ratings").

e.g. (main) rating classes of Standard&Poor's:

AAA, AA, A, BBB, BB, BB, CCC, CC, C, D(efault)

- *Note:* Even when applying statistical methods the classification of obligors into rating classes is made (for internal communication, standardization, comparability).
- The ratings are the basis for the estimation of the probability of default.

Probability of Default based on Ratings

- For these rating classes the probability of default has to be estimated based on the observed defaults.
- In the following sections the methods for this estimation will be shown. Basically two methods are available:
 - Discrete transition matrices based on the cohort method
 - Continuous transition matrices

- The most simple method to estimate probabilities of default is the cohort method:
 - All obligors have to be classified into cohorts (rating classes)
 - Each rating class should represent a homogeneous group of obligors, i.e. a rating class should consist out of obligors where one assumes that they have identical probabilities of default.
 - Given a certain period of time (e.g. one year interval) the PD of a rating class is estimated as follows:

$$PD = \frac{\text{number of defaults in the chosen time period}}{\text{number of obligors at the beginning of the time period}}$$

- The cohort method uses to critical assumptions.
 1. Assumption: No obligor leaves the portfolio in the chosen time period (e.g. because his loan matures).
 2. Assumption: New obligors that become part of the portfolio during the time period are assumed to have the same probability of default as the old obligors.

Note: If these assumptions are not fulfilled the estimates of the PD will be biased.

- Example: A bank has a rating system with two rating classes A and B. The rating class A represents obligors of “good” credit quality and rating class B represents obligors of “bad” credit quality. During a time period of two years the following information is collected:

	rating class A	rating class B
number of obligors	21.000	36.000
number of defaults after one year	209	520
number of defaults after two years	387	1.260

- Estimate the PD for a time period of one and two years in the rating class A and B.

	rating class A	rating class B
one year PD	$209 / 21.000 = 0,995\%$	$520 / 36.000 = 1,444\%$
two years PD	$387 / 21.000 = 1,843\%$	$1.260 / 36.000 = 3,5\%$

- Alternatively, a better estimate for the one year PD could be calculated based on the whole two year time period, e.g. time weighted:

$$\text{one year PD of rating class A} = 0.5 \cdot \frac{209}{21000} + 0.5 \cdot \frac{178}{20791} = 0.926\%$$

$$\text{one year PD of rating class B} = 0.5 \cdot \frac{520}{36000} + 0.5 \cdot \frac{740}{35480} = 1.765\%$$

- Due to the scarcity of data and the critical assumptions of the cohort method normally only one year PDs are estimated.
- This is sufficient for regulatory purposes. But for other tasks (e.g. pricing) PDs estimated for longer time periods are needed. These PDs can be estimated using transition matrices.

- Question: Is the information of one year PDs sufficient for the calculation of multi year PDs?

1. Scenario: Assume that we have a rating system with only one rating class with a one year PD of pd .

$$\begin{array}{lcl} \text{two year PD of the rating class} & = & pd + (1 - pd) \cdot pd \\ & & \text{1.year} \quad \quad \text{2. year} \end{array}$$

→ In this simple scenario the knowledge of one year PDs is sufficient.

2. Scenario: The rating system has two rating classes (A and B) with one year PDs of pd_A and pd_B .

The two year PD for an obligor of rating class A is now also dependent on the probability that this obligor will be in rating class B after one year. Lets call the one year probability to migrate from A to B $p_{A,B}$

$$\text{two year PD of the rating class A} = \underset{\text{1.year}}{pd_A} + p_{A,B} \cdot \underset{\text{2.year}}{pd_B} + (1 - pd_A - p_{A,B}) \cdot pd_A$$

→ To estimate multi year PDs all one year migration probabilities must be available! These probabilities are estimated using transition matrices.

Discrete Transition Matrix

- A discrete transition matrix contains the one year probabilities to migrate from one rating class to another rating class (including the default rating class).
- The probability to migrate from one rating class to another can be calculated by using the cohort method. Probability to migrate from rating class i to rating class j:

$$\frac{\text{number of migrations from rating class i to rating class j}}{\text{number of obligors in rating class i at the beginning of the year}}$$

- The assumptions of the cohort method still have to be fulfilled.

Discrete Transition Matrix

- Example: A bank has two rating classes (A and B) where A represents „good“ credit quality and B represents „bad“ credit quality. The following information is available after one year.

	rating class A	rating class B
number of obligors	21.000	36.000
migrations to rating class B	3.150	-
migrations to rating class A	-	1.800
defaults	210	720

Calculate the one year transition matrix for this rating system.

Discrete Transition Matrix

Solution:

probability to stay in A: $(21000 - 3150 - 210) / 21000 = 0.84$

probability to migrate from A to B: $3150 / 21000 = 0.15$

probability to migrate from A to Default: $210 / 21000 = 0.01$

probability to stay in B: $(36000 - 1800 - 720) / 36000 = 0.93$

probability to migrate from B to A: $1800 / 36000 = 0.05$

probability to migrate from B to Default : $720 / 36000 = 0.02$

one year transition matrix:

From / To	A	B	Default
A	84%	15%	1%
B	5%	93%	2%
Default	0%	0%	100%

Discrete Transition Matrix

- The one year PDs can be directly taken from a one year transition matrix.
- More important, a multi year transition matrix can easily be estimated from a one year transition matrix by multiplying the transition matrix with itself.
- Calculation of a two year transition matrix with n rating classes (including default):

$$\begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{pmatrix} \cdot \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{pmatrix} \text{ or writing } \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{pmatrix}^2$$

p_{ij} one year probability to migrate from i to j

Discrete Transition Matrix

- Calculate the two year transition matrix for our example:

$$\begin{pmatrix} 0.84 & 0.15 & 0.01 \\ 0.05 & 0.93 & 0.02 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0.84 & 0.15 & 0.01 \\ 0.05 & 0.93 & 0.02 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.7131 & 0.2655 & 0.0214 \\ 0.0885 & 0.8724 & 0.0391 \\ 0 & 0 & 1 \end{pmatrix}$$

Discrete Transition Matrix

- Calculation of a k-year transition matrix with n rating classes (including default):

$$\begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{pmatrix}^k$$

Discrete Transition Matrix

- 2-year discrete transition matrix based on Standard & Poor's:

From / To	AAA	AA	A	BBB	BB	B	CCC/C	Default
AAA	84.83%	12.98%	1.71%	0.34%	0.12%	0.01%	0.00%	0.002%
AA	1.14%	82.71%	14.19%	1.52%	0.17%	0.20%	0.04%	0.04%
A	0.11%	3.82%	83.90%	10.50%	1.06%	0.37%	0.08%	0.15%
BBB	0.06%	0.47%	7.43%	80.40%	8.39%	1.90%	0.44%	0.91%
BB	0.06%	0.17%	0.96%	9.60%	70.03%	13.67%	1.92%	3.59%
B	0.01%	0.15%	0.53%	0.93%	9.02%	68.88%	6.71%	13.78%
CCC/C	0.15%	0.02%	0.48%	0.96%	2.71%	14.38%	28.42%	52.89%
Default	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%

Discrete Transition Matrix

- 10-year discrete transition matrix based on Standard & Poor's:

From / To	AAA	AA	A	BBB^k	BB	B	CCC/C	Default
AAA	44.85%	33.03%	15.75%	4.44%	1.01%	0.49%	0.09%	0.333%
AA	2.96%	42.86%	36.44%	12.29%	2.65%	1.37%	0.24%	1.16%
A	0.59%	10.01%	49.57%	26.36%	6.78%	3.16%	0.55%	2.98%
BBB	0.28%	2.90%	18.94%	42.18%	16.14%	8.49%	1.39%	9.69%
BB	0.19%	1.06%	6.03%	18.31%	25.04%	19.23%	2.95%	27.19%
B	0.09%	0.56%	2.35%	5.76%	12.56%	22.01%	3.41%	53.28%
CCC/C	0.14%	0.28%	1.16%	2.41%	4.23%	7.35%	1.27%	83.17%
Default	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%

- Based on the discrete transition matrix one year and multi year default probabilities can be estimated. Two drawbacks remain:
 - The two assumptions must be fulfilled.
 - Probabilities can only be estimated for whole years
- For scenarios where this is a problem, continuous transition matrices can be used.
- The main advantage of this technique is that it uses the whole rating information and not only the information of the beginning and the end of a time period. Thus this technique can also integrate the information that obligors leave or enter the portfolio.
- Also „realistic“ PDs can be estimated for rating classes with no observed defaults.

- Method: Lando and Skødeberg [2002]
- Models PDs with a time homogenous Markov chain in continuous time.
- The transition matrix is modelled using a generator matrix Λ . The elements of this matrix are the intensities λ_{ij} to migrate between rating class i and j .
- The intensities of a migration contain the information of the probability of a migration in the next infinitesimal time step Δt :

$$\lambda_{ij} \cdot \Delta t = p_{ij}(\Delta t)$$

- This relation allows to estimate PDs for any time period.

- Estimation of the generator matrix Λ :
 - In this model, the observed rating changes and defaults are used to estimate the generator matrix, instead of directly estimating the transition matrix.
 - The estimator (Maximum-Likelihood) for the intensities are given as:

$$\lambda_{ij} = \frac{N_{ij}}{\int Y_i(s) ds} \quad i \neq j \quad \text{und} \quad \lambda_{ii} = -\sum_{j \neq i} \lambda_{ij}$$

N_{ij} Number of migrations from rating class i to j

$Y_i(s)$ Number of obligors in rating class i at time s

- There is a simple intuition behind these estimations:
- The numerator (N_{ij}) counts the migrations from rating class i to j (as in the discrete case).
- The denominator counts the overall „obligor-years“ observed in rating class i .
- Thus, the time spent in the rating class by an obligor is precisely measured.
- Consider the following example.

Continuous Transition Matrix

- Example: A bank has a rating system with two rating classes (A und B). Rating class A is considered for „good“ credit quality and B for „bad“ credit quality. The following data is observed within one year:

There are 10 obligors in each rating class at the beginning of the year.

One „A“ obligor migrates to rating class „B“ after one month and stays there.

One „B“ obligor migrates to rating class „A“ after two months and stays there.

One „B“ obligor defaults after six months

Estimate the generator matrix.

Continuous Transition Matrix

- Solution:

Intensity to migrate from „A“ to „B“:

$$\lambda_{AB} = \frac{N_{AB}}{\int Y_A(s) ds} = \frac{1}{10 \cdot \frac{1}{12} + 9 \cdot \frac{1}{12} + 10 \cdot \frac{10}{12}} = 0.10084$$

Intensity to migrate from „A“ to Default „D“:

$$\lambda_{AD} = \frac{N_{AD}}{\int Y_A(s) ds} = \frac{0}{10 \cdot \frac{1}{12} + 9 \cdot \frac{1}{12} + 10 \cdot \frac{10}{12}} = 0$$

$$\rightarrow \lambda_{AA} = -\sum_{j \neq A} \lambda_{Aj} = -\lambda_{AB} - \lambda_{AD} = -0.10084$$

- Solution:

Intensity to migrate from „B“ to „A“:

$$\lambda_{BA} = \frac{N_{BA}}{\int Y_B(s) ds} = \frac{1}{10 \cdot \frac{1}{12} + 11 \cdot \frac{1}{12} + 10 \cdot \frac{4}{12} + 9 \cdot \frac{6}{12}} = 0.10435$$

Intensity to migrate from „B“ to Default „D“:

$$\lambda_{BD} = \frac{N_{BD}}{\int Y_B(s) ds} = \frac{1}{10 \cdot \frac{1}{12} + 11 \cdot \frac{1}{12} + 10 \cdot \frac{4}{12} + 9 \cdot \frac{6}{12}} = 0.10435$$

$$\rightarrow \lambda_{BB} = -\sum_{j \neq B} \lambda_{Bj} = -\lambda_{BA} - \lambda_{BD} = -0.2087$$

- Solution:

Intensities for migrating from Default classes are set to zero:

$$\lambda_{DA} = \lambda_{DB} = \lambda_{DD} = 0$$

Generator matrix:

$$\Lambda = \begin{pmatrix} -0.10084 & 0.10084 & 0 \\ 0.10435 & -0.2087 & 0.10435 \\ 0 & 0 & 0 \end{pmatrix}$$

- Estimation of the transition matrix:
 - The generator matrix can be used to estimate PDs for any time period Δt .
 - The transition matrix $W(\Delta t)$ is estimated using

$$W(\Delta t) = \exp(\Lambda \cdot \Delta t) = \sum_{k=0}^{\infty} \Lambda^k \cdot \frac{\Delta t^k}{k!}$$

- This matrix contains the migration and default probabilities.

Continuous Transition Matrix

- **Example:** Use the generator matrix from the previous example to estimate the continuous transition matrix for a time period of one year (Δt). Use the first four terms of the sum for this exercise.

Generator matrix:

$$\Lambda = \begin{pmatrix} -0.10084 & 0.10084 & 0 \\ 0.10435 & -0.2087 & 0.10435 \\ 0 & 0 & 0 \end{pmatrix}$$

Solution:

1. term:

$$\Lambda^0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \frac{\Delta t^0}{0!} = 1 \quad \Lambda^0 \cdot \frac{\Delta t^0}{0!} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Continuous Transition Matrix

- 2. term:

$$\Lambda^1 = \begin{pmatrix} -0.10084 & 0.10084 & 0 \\ 0.10435 & -0.2087 & 0.10435 \\ 0 & 0 & 0 \end{pmatrix} \quad \frac{\Delta t^1}{1!} = 1 \quad \Lambda^1 \cdot \frac{\Delta t^1}{1!} = \begin{pmatrix} -0.10084 & 0.10084 & 0 \\ 0.10435 & -0.2087 & 0.10435 \\ 0 & 0 & 0 \end{pmatrix}$$

- 3. term:

$$\Lambda^2 = \begin{pmatrix} 0.02069 & -0.03121 & 0.01052 \\ -0.03230 & 0.05407 & -0.02178 \\ 0 & 0 & 0 \end{pmatrix} \quad \frac{\Delta t^2}{2!} = 0.5 \quad \Lambda^2 \cdot \frac{\Delta t^2}{2!} = \begin{pmatrix} 0.010345 & -0.015605 & 0.00526 \\ -0.01645 & 0.02704 & -0.01089 \\ 0 & 0 & 0 \end{pmatrix}$$

- 4. term:

$$\Lambda^3 = \begin{pmatrix} -0.00534 & 0.0086 & -0.00326 \\ 0.0089 & -0.01454 & 0.00564 \\ 0 & 0 & 0 \end{pmatrix} \quad \frac{\Delta t^3}{3!} = 0.16 \quad \Lambda^3 \cdot \frac{\Delta t^3}{3!} = \begin{pmatrix} -0.00089 & 0.00143 & -0.000543 \\ 0.001483 & -0.002423 & 0.00094 \\ 0 & 0 & 0 \end{pmatrix}$$

- Transition matrix:

$$\Lambda^0 \cdot \frac{\Delta t^0}{0!} + \Lambda^1 \cdot \frac{\Delta t^1}{1!} + \Lambda^2 \cdot \frac{\Delta t^2}{2!} + \Lambda^3 \cdot \frac{\Delta t^3}{3!} = \begin{pmatrix} 0.90862 & 0.08667 & 0.00472 \\ 0.08938 & 0.81592 & 0.09440 \\ 0 & 0 & 1 \end{pmatrix}$$

→ one-year PD for „A”: 0.472%

→ one-year PD for „B”: 9.44%

Continuous Transition Matrix

- Example: Estimate the discrete transition matrix using the information of the previous examples and compare the results to the continuous transition matrix:

Solution:

Probability to stay in A:	$(10 - 1) / 10 = 0.9$
Probability to migrate from A to B:	$1 / 10 = 0.1$
Probability to migrate from A to Default:	$0 / 10 = 0$
Probability to stay in B:	$(10 - 1 - 1) / 10 = 0.8$
Probability to migrate from B to A:	$1 / 10 = 0.1$
Probability to migrate from B to Default:	$1 / 10 = 0.1$

Continuous Transition Matrix

- One-year discrete transition matrix:

From / To	A	B	Default
A	90%	10%	0%
B	10%	80%	10%
Default	0%	0%	100%

- One-year continuous transition matrix:

From / To	A	B	Default
A	90.86%	8.67%	0.47%
B	8.97%	81.59%	9.44%
Default	0%	0%	100%

Continuous Transition Matrix

- Example: Estimate the continuous transition matrix for a time period of 0.5 and 2.5 years based on the generator matrix using the first four terms of the sum:

- Solution for 0.5 years:

From / To	A	B	Default
A	95.21%	4.67%	0.12%
B	4.83%	90.21%	4.96%
Default	0%	0%	100%

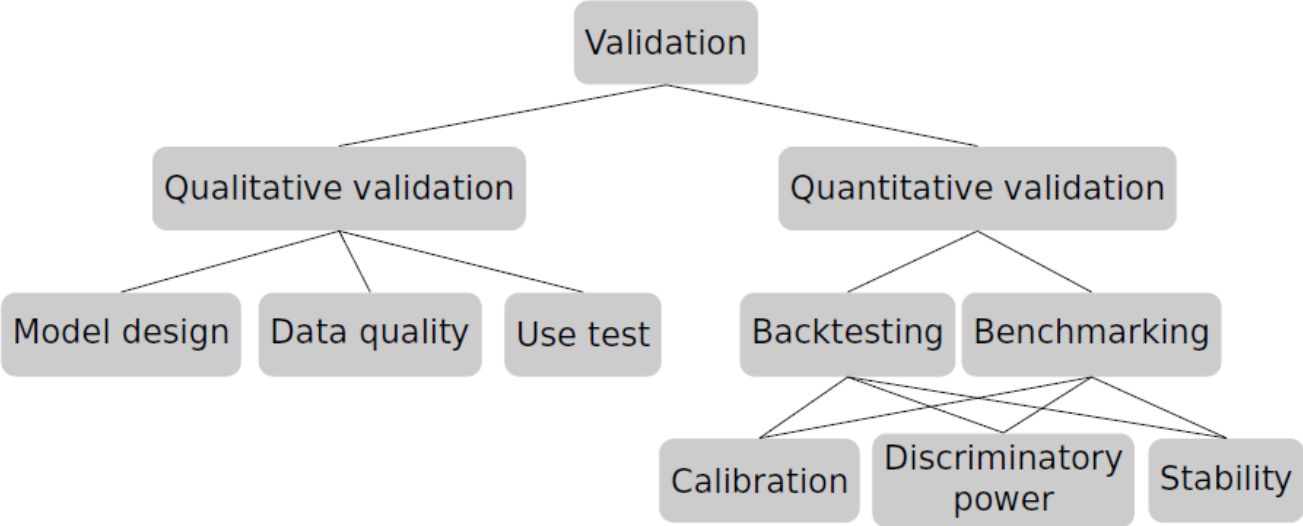
- Solution for 2.5 years:

From / To	A	B	Default
A	79.86%	17.70%	2.44%
B	18.31%	60.94%	20.75%
Default	0%	0%	100%

PD Validation

- Validation is a crucial process within banks to assure that the PD estimates are sound.
- The CRR contains various rules / requirements concerning validation. In addition, guidelines and technical standards provide additional information.
- CRR Article 185 (Validation of internal estimates): Institutions shall validate their internal estimates subject to the following requirements:
 - (a) institutions shall have robust systems in place to validate the accuracy and consistency of rating systems, processes, and the estimation of all relevant risk parameters. . . .
 - (b)) institutions shall regularly compare realized default rates with estimated PDs for each grade and, where realized default rates are outside the expected range for that grade, institutions shall specifically analyze the reasons for the deviation. . . .
 - (c) institutions shall also use other quantitative validation tools and comparisons with relevant external data sources. . . .

- CRR Article 188 (Validation and documentation): Institutions shall have robust systems in place to validate the accuracy and consistency of their internal models and modelling processes. All material elements of the internal models and the modelling process and validation shall be documented.
- (a) institutions shall use the internal validation process to assess the performance of its internal models and processes in a consistent and meaningful way;
- (f) the internal model and the modelling process shall be documented, including the responsibilities of parties involved in the modelling, and the model approval and model review processes.



- The quantitative validation is focused on statistics and statistical test to evaluate the PD estimated.
- Often it is not possible to directly reject or approve a rating process based on these quantitative results.
- Therefore, also qualitative criteria are used to validate the rating process.
- These qualitative tests are focused on:
 - Model design analyzing the rating documentation
 - Data quality of the characteristics, ratings and defaults
 - Use test focusing on processes where ratings are used.

- Model design:
 - The design of the rating process is very important in case no quantitative validation is possible.
 - The documentation of the rating model is analyzed, to understand whether the rating process is clearly described and transparent.
 - It has to be clear what characteristics have been chosen and why. Their impact has to be reasonable in economic terms. The rating method has to be chosen in accordance with the data quantity and the chosen characteristics.
- Data quality:
 - The most important issue is whether characteristics, ratings and defaults are available for a reasonable time period and whether this information can be accessed directly.

- Use test:
 - The usage of ratings within the processes and the reporting of banks are evaluated.
 - Are ratings used in the:
 - Pricing
 - Loan granting
 - Limit system
 - Capital allocation
 - If ratings are only used for isolated purposes (e.g. only for minimum capital requirements), the reason could be the low quality of the ratings.
 - An additional point is whether the employees have the qualification to understand and interpret ratings.

- Most important criteria:
 - Calibration
 - Discriminatory power
 - Stability
- Calibration: agreement of predicted probabilities of default and realized default rates
- Discriminatory power: ability to differentiate “good” and “bad” obligors
- Stability: robustness with respect to changes (time, portfolio, surrounding conditions, , . . .)
- Backtesting: comparison of predictions and realizations
- Benchmarking: comparison with different models/methods

Quantitative Validation: Calibration

- Example from the OeNB/FMA Guidelines on Credit Risk Management: Numbers of obligors (n) and defaults (d) and corresponding model PDs (p) for a rating system with 10 rating classes:

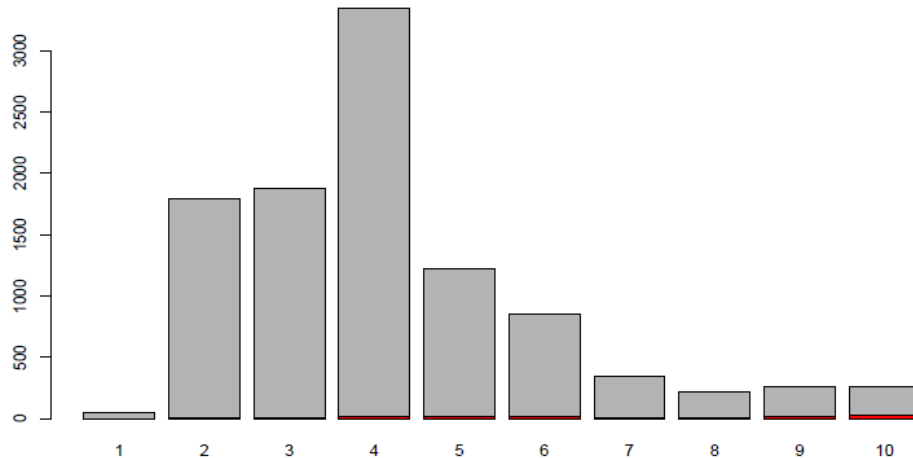
	n	d	p
1	50	0	0.0005
2	1788	3	0.0010
3	1876	6	0.0025
4	3345	15	0.0050
5	1223	15	0.0075
6	856	17	0.0130
7	342	12	0.0200
8	214	9	0.0340
9	265	22	0.0600
10	257	33	0.1000

- When starting quantitative validation, take a look at:
 - Structure of rating classes (Number of classes and corresponding PDs)
 - Distribution of obligors according to rating class:
 - Do we have a portfolio of good or bad obligors?
 - Are there concentrations in specific rating classes?
 - Is there a sufficient number of rating classes?
 - Numbers of obligors and defaults:
 - Are there sufficiently many observations?
 - What is the default rate for the portfolio?

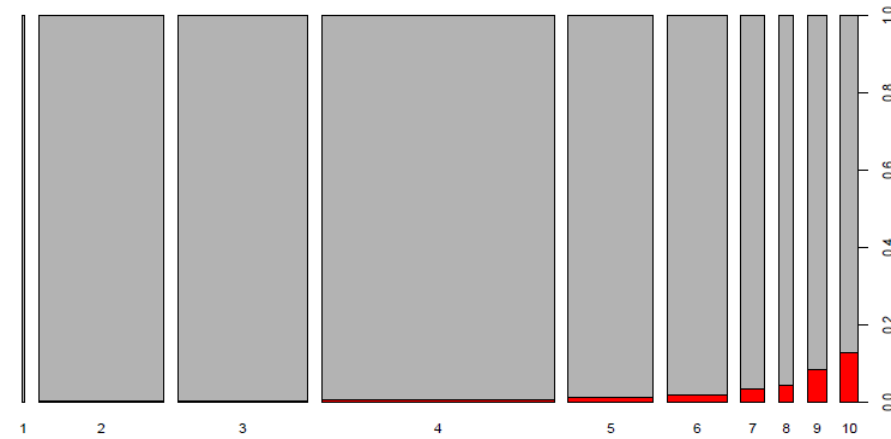
Quantitative Validation: Calibration

- Total number of obligors: 10216
- Total number of defaults: 132
- Default rate for the portfolio: 0.0129

Barplots

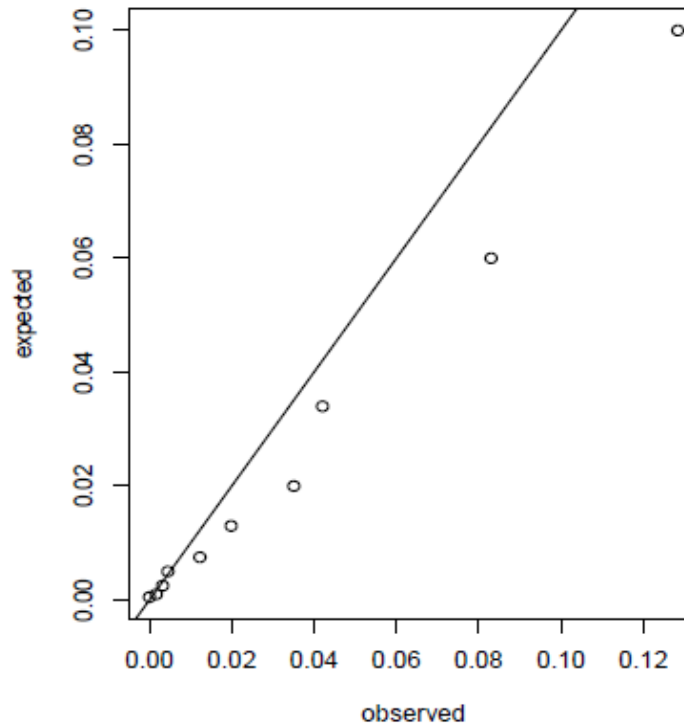


Spineplots



Quantitative Validation: Calibration

- Comparison of expected and observed default rates, on the PD scale:



- How “significant” are the differences between observed and expected/estimated default rates?
- Need appropriate statistical tests (hypothesis tests).
- Need appropriate probabilistic models for the default data generating process.
- Simplest solution: Binomial distribution
 - Number of defaults in a rating class has a binomial distribution with n = number of obligors in rating class and p = expected probability of default (PD):

$$\mathbb{P}(D = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Note: this assumes that defaults occur **independently** (expected number of defaults = $n \cdot p$)

- Example: A rating class has 20 obligors and an estimated PD of 5%. What is the probability of observing more than 4 defaults?

Solution:

$$\mathbb{P}(D = 0) = \frac{20!}{0!20!} 0.05^0 0.95^{20} = 35.85 \%$$

$$\mathbb{P}(D = 1) = \frac{20!}{1!19!} 0.05^1 0.95^{19} = 37.74 \%$$

$$\mathbb{P}(D = 2) = \frac{20!}{2!18!} 0.05^2 0.95^{18} = 18.87 \%$$

$$\mathbb{P}(D = 3) = \frac{20!}{3!17!} 0.05^3 0.95^{17} = 5.96 \%$$

$$\mathbb{P}(D = 4) = \frac{20!}{4!16!} 0.05^4 0.95^{16} = 1.33 \%$$

Therefore:

$$\mathbb{P}(D \leq 4) = (35.85 + 37.74 + 18.87 + 5.96 + 1.33) \% = 99.75 \%$$

from which

$$\mathbb{P}(D > 4) = 1 - \mathbb{P}(D \leq 4) = 0.25 \%$$

- Example: A rating class has 20 obligors and an estimated PD of 5%. What is the probability of observing more than 4 defaults?

With R:

```
R> ## P(D <= 4):  
R> pbinom(4, 20, 0.05)  
  
[1] 0.9974261  
  
R> ## P(D > 4):  
R> 1 - pbinom(4, 20, 0.05)  
  
[1] 0.00257394
```

- To test whether the expected/modeled PD p_0 is appropriate, we can perform a hypothesis test of

$$H_0 : p = p_0$$

- With the alternative that PDs are correct:

$$H_A : p \neq p_0$$

two-sided alternative (i.e., PDs are not estimated correctly)

- CRR Article 174(a) says “The model shall not have material biases”.

- Rating class 10 showed the largest difference between observed and expected default rates:

Exact binomial test

```
data: d and n
number of successes = 33, number of trials = 257,
p-value = 0.1442
alternative hypothesis: true probability of success is not equal to 0.1
95 percent confidence interval:
 0.09006213 0.17557866
sample estimates:
probability of success
      0.1284047
```

- Still not statistically significant at the 10% level.

Quantitative Validation: Calibration

- All rating classes are tested:

	n	d	p	exact
1	50	0	0.0005	1.0000
2	1788	3	0.0010	0.2661
3	1876	6	0.0025	0.4827
4	3345	15	0.0050	0.8058
5	1223	15	0.0075	0.0652
6	856	17	0.0130	0.0934
7	342	12	0.0200	0.0527
8	214	9	0.0340	0.4505
9	265	22	0.0600	0.1194
10	257	33	0.1000	0.1442

- Another idea is to use omnibus tests which simultaneously test calibration for all rating classes.
- Hosmer-Lemeshow test: One assumes that all defaults occur independently, so that the numbers of defaults $D_1, \dots, D_K$ are independent and have binomial distributions with parameters n_k and (unknown) p_k , and tests

$$H_0 : p_k = p_{0,k}, k = 1, \dots, K; \quad H_A : \text{the opposite of } H_0$$

- If all $n_k \cdot p_{0,k}$ are “large enough”, then (under H_0) the test statistic can be approximated by chi-squared distribution with K degrees of freedom (in R: prop.test).

- Result from the Hosmer-Lemeshow test:

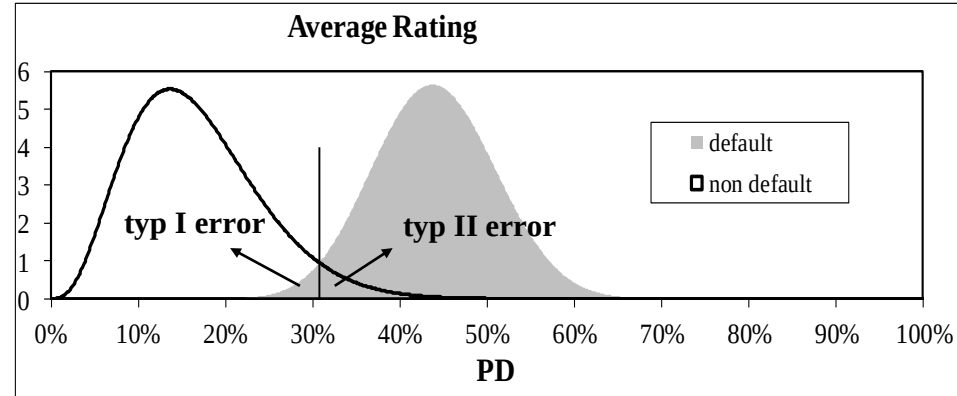
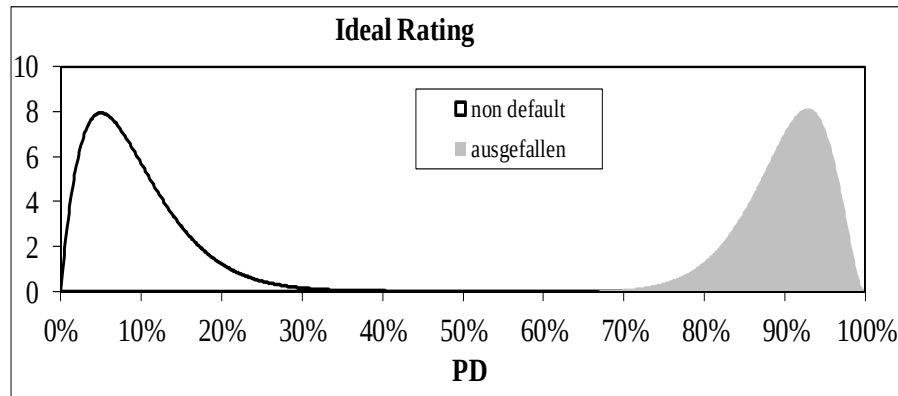
10-sample test for given proportions without
continuity correction

```
data:  d out of n, null probabilities p
X-squared = 17.451, df = 10, p-value = 0.06496
alternative hypothesis: two.sided
null values:
  prop 1  prop 2  prop 3  prop 4  prop 5  prop 6  prop 7
  0.0005  0.0010  0.0025  0.0050  0.0075  0.0130  0.0200
  prop 8  prop 9  prop 10
  0.0340  0.0600  0.1000
sample estimates:
      prop 1      prop 2      prop 3      prop 4      prop 5
0.0000000000 0.001677852 0.003198294 0.004484305 0.012264922
      prop 6      prop 7      prop 8      prop 9      prop 10
0.019859813 0.035087719 0.042056075 0.083018868 0.128404669
```

- In addition, regulatory documents recommend to use the Jeffreys test.
- This is a Bayesian hypothesis test using the Jeffreys prior for the unknown PD p_k .
- This test produces more conservative p-values, but still assumes independence of defaults.
- All these tests are assumed to be „conservative“, as in most credit risk models defaults have to be correlated to produce risk. Test based on such assumptions need to simulate confidence intervals.

Quantitative Validation: Discriminatory power

- Ability of a rating system to differentiate “good” and “bad” obligors upfront.
- Fundamental problem: good versus bad is not the same as non-defaulted versus defaulted.
- Need appropriate measures for the ordinal association between observed and expected default rates.



Quantitative Validation: Discriminatory power

- Brier Score:
 - Brier Score is a simple statistic to measure discriminatory power.
 - The Score is estimated based on n obligors based on the mean squared error:

$$\text{Score} = \frac{1}{n} \cdot \sum_{i=1}^n (\text{PD}_i - Y_i)^2$$

Y_i 1... obligor i defaulted, 0... obligor i no default

- The ideal score is 0.

Quantitative Validation: Discriminatory power

- Cumulative Accuracy Profile (CAP):
 - The CAP is based on the following idea.
 - Suppose we order obligors in increasing expected creditworthiness. (careful: i.e., in decreasing PDs!)
 - “Ideally” only obligors with the smallest creditworthiness default.
 - “Naively” (if the expectations are just random guessing, and perform no risk differentiation at all) defaults are distributed randomly across the creditworthiness classes.
 - In reality, we can look at the numbers of defaults as a function of creditworthiness. I.e., we look at the relative frequencies of defaults as a function of the relative frequencies of grouping according to increasing creditworthiness.

Quantitative Validation: Discriminatory power

- Accuracy Ratio (or Gini coefficient, PowerStat):
 - The accuracy ratio gives a measure of discriminatory power based on the Cumulative Accuracy Profile.
 - Write B for the area under the CAP curve and above the first median.
 - We compare this with the area A under the CAP curve for the “ideal rating” (where all defaults occur in the worst rating class) and above the first median.

$$AR = B/A.$$

- For the ideal rating: $AR = 1$, for random guessing: $AR = 0$.
- Idea: methods/ratings with more discriminatory power have higher accuracy ratios.

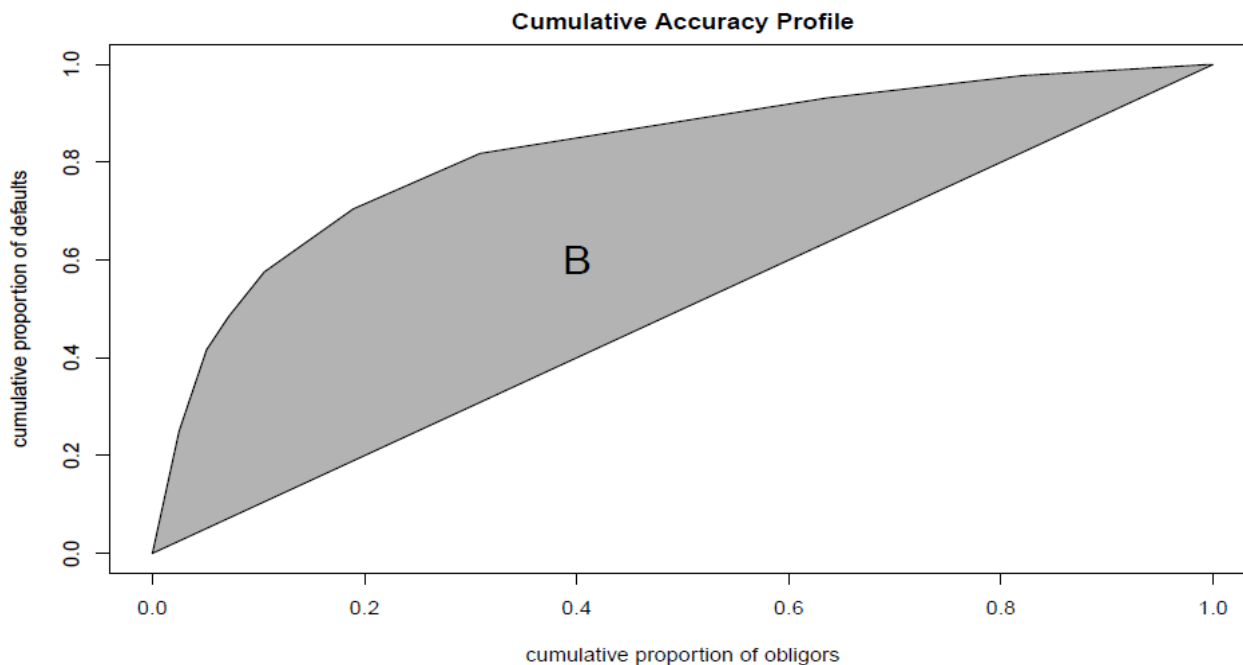
Quantitative Validation: Discriminatory power

- Using the previous example, We augment the data with the cumulative proportions of obligors and defaults with at most equal creditworthiness:

	n	d	p	x	y
1	50	0	0.0005	1.0000	1.0000
2	1788	3	0.0010	0.9951	1.0000
3	1876	6	0.0025	0.8201	0.9773
4	3345	15	0.0050	0.6365	0.9318
5	1223	15	0.0075	0.3090	0.8182
6	856	17	0.0130	0.1893	0.7045
7	342	12	0.0200	0.1055	0.5758
8	214	9	0.0340	0.0720	0.4848
9	265	22	0.0600	0.0511	0.4167
10	257	33	0.1000	0.0252	0.2500

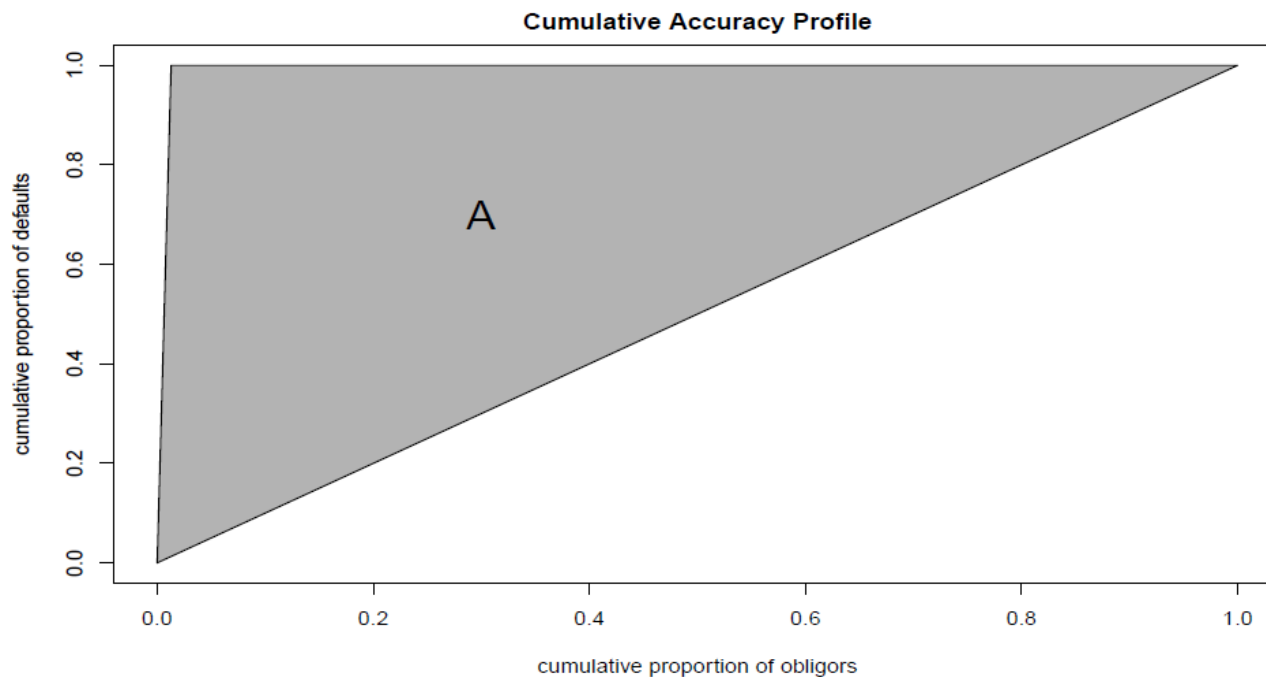
Quantitative Validation: Discriminatory power

- CAP for the actual rating: $B = 0.3235$.



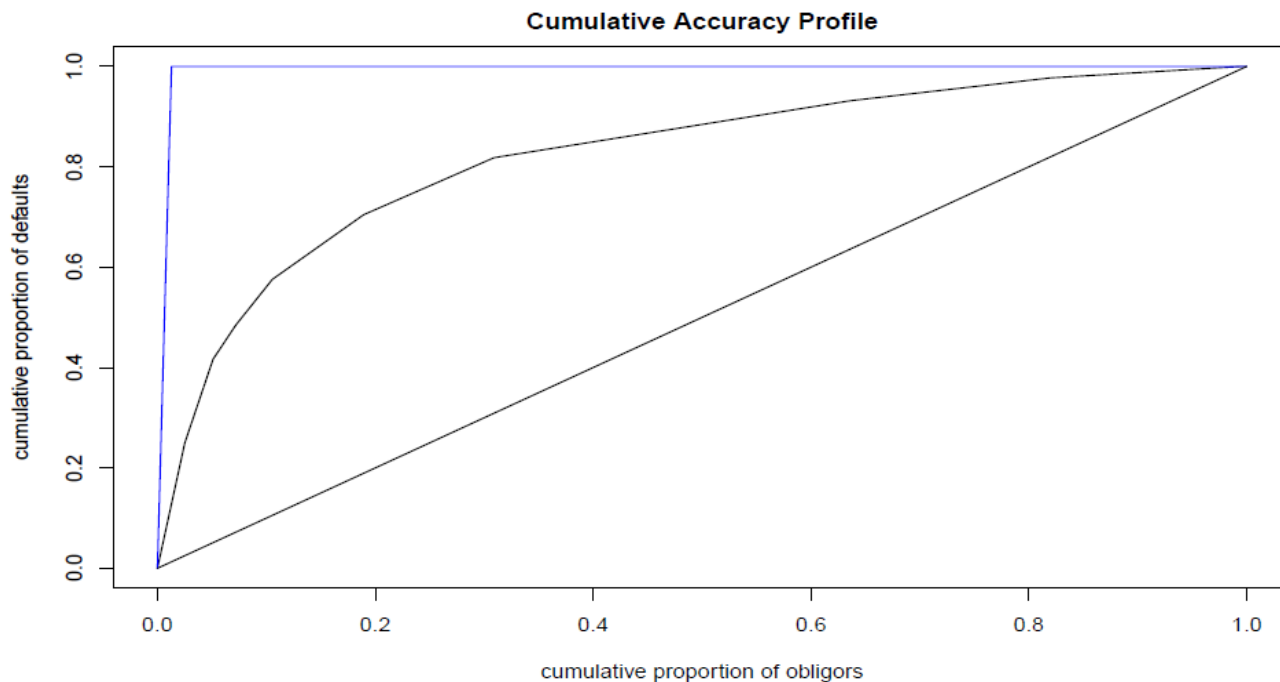
Quantitative Validation: Discriminatory power

- CAP for the ideal rating: $A = 0.4935$.



Quantitative Validation: Discriminatory power

- Combined CAPs: $AR = 0.6554 (= 0.3235 / 0.4935)$



Quantitative Validation: Discriminatory power

- Does this indicate high discriminatory power?
- Traditionally (e.g., in the guidelines), one used thresholds to indicate “good” (“sufficient”) discriminatory power.
- But the ARs depend on the composition of the portfolio and the granularity of the rating system employed, and hence cannot really meaningfully be compared across portfolios or systems.
- See, e.g., Lingo & Winkler (2008), “Discriminatory power: an obsolete validation criterion?”, Journal of Risk Model Validation.
- In the new ECB guidelines the AR is observed over time to analyze whether the AR reduced over time. This brings us to stability.

- CRR Article 174 (d): the institution shall have a regular cycle of model validation that includes monitoring of model performance and stability;
- But what precisely does “stability” mean?
- One possibility is to observe calibration and discriminatory power over time. Reduced performance after the first years of development is a sign of poor performance (e.g. overfitting).
- For some measures, e.g., AR, ECB guidelines offer test statistics.
- In addition, the (changes in the) distribution of obligors across rating classes can be observed: migrations and concentrations.
- The ECB guidelines offers test statistics as well.

- In many situations the quantitative validation is not possible as the number of observations is simply too low, e.g.:
 - Number of obligors too low
 - Number of defaults too low
 - Rating system is newly developed, no time series
- In this cases, it might be possible to compare the results of the rating system to other rating results (e.g., external ratings).
- If the different rating source use or a mapped on the same rating scale, then the starting point of a benchmarking analysis is the cross-tabulation of obligors.

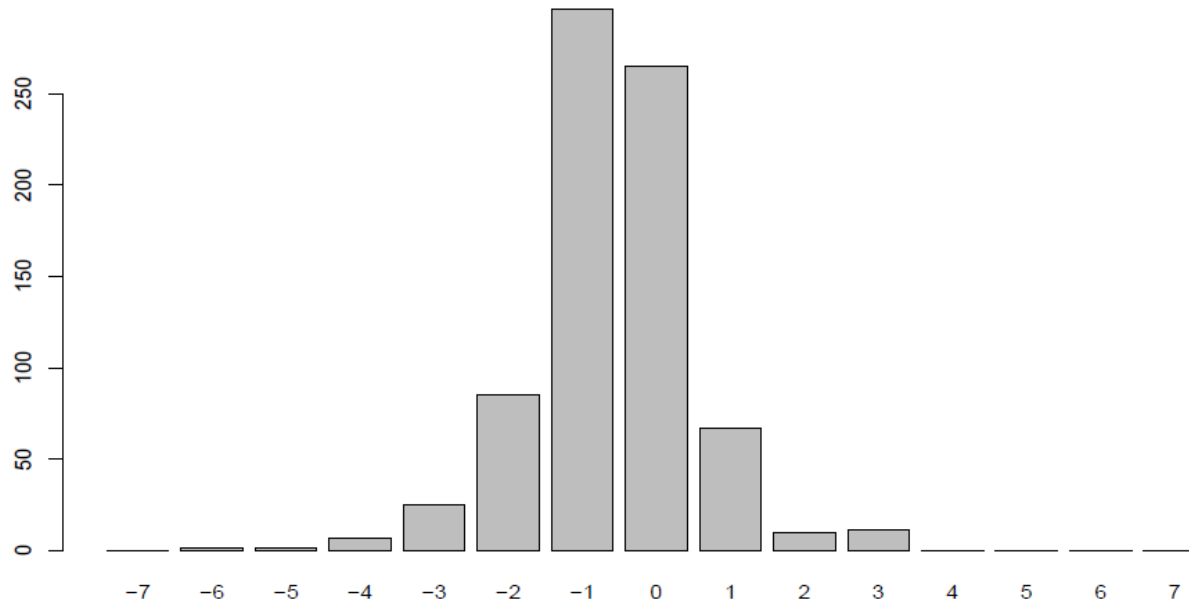
Quantitative Validation: Benchmarking

- The supervisor might be in the favorable position to receive the ratings of different banks, when they report to a central credit register:

			Bank2	Bank1								
				Rating	1	2	3	4	5	6	7	8
		ce		1	0	51	12	1	0	0	1	0
	Bank1	Bank		2	0	12	93	30	10	0	0	0
Bank1		768		3	0	6	104	66	27	10	0	1
Bank2	768			4	0	0	28	83	70	14	3	7
Bank3	1159	498		5	0	0	2	19	41	13	1	1
Bank4	728	280		6	0	0	1	4	10	4	1	1
Bank5	410	214		7	0	0	0	3	2	4	0	2
				8	0	0	0	0	7	2	0	21

Quantitative Validation: Benchmarking

- Deviations can be analyzed:



- Information in rating deviations can be evaluated with suitable measures.
- Once upon a time . . . there was a research cooperation between WU and OeNB suggesting the following aspects and measures of “proximity”:
 - Agreement (amount of deviation): Cohen´s Kappa (κ)
 - Bias (direction of deviation): Theta (θ)
 - Association: Tau X (τ_x)
- References:
 - Hornik, Jankowitsch, Lingo, Pichler und Winkler (2007)
 - Hornik, Jankowitsch, Lingo, Pichler und Winkler (2010)

- The three measures:

$$\kappa = \frac{\sum_{j=1}^R \sum_{i=1}^R w_{ij} p_{ij} - \sum_{j=1}^R \sum_{i=1}^R w_{ij} p_i \cdot p_j}{1 - \sum_{j=1}^R \sum_{i=1}^R w_{ij} p_i \cdot p_j} \quad \text{mit} \quad w_{ij} = 1 - \left(\frac{i-j}{R-1} \right)^2$$

$$\theta = \sum_{j=1}^R \sum_{i=1}^R \frac{i-j}{R-1} p_{ij}$$

$$\tau_x = 1 - \frac{\sum_{u=1}^N \sum_{v=1}^N |a_{uv} - b_{uv}|}{N(N-1)}$$

N ... number co-ratings
 R ... number of rating classes
 u, v ... obligor
 i, j ... rating class
 p ... rel. frequency of observations
 a, b ... 1, if rating of u better than rating of v , otherwise 0

Quantitative Validation: Benchmarking

- Benchmarking example:

RATINGS		<i>Bank 1</i>	<i>Bank 2</i>
O b l i g o r	1	1	2
	2	2	3
	3	3	4
	4	4	5
	5	5	6
	6	6	7
	7	7	8

$$\tau_x = 1 \quad \kappa = 0.889 \quad \theta = 0.143$$

Exposure at Default (EAD)

Introduction – Exposure at Default

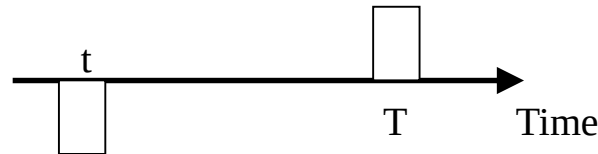
- Exposure at Default: *EAD* represents the amount outstanding at the default of an obligor/obligation.
- The amount outstanding represents the credit exposure that has to be repaid plus interest rates that were due at default.
- The estimation of the expected *EAD* has to be tailored to the transaction type and shows various degrees of complexity.
- The parameter *EAD* represent the prerequisite to estimate the loss rate (Loss Given Default).

Introduction – Exposure at Default

- The basis for estimating the EAD is the current book value of a loan (nominal value of exposure).
- Furthermore, potential changes of this amount have to be considered. This potential will depend on the transaction type.
- These changes can depend on the decision of the customer, the realization of certain event, or the development of market prices.
- In general, we can distinguish the following transaction types for the estimation of the EAD.

EAD – Transaction Types

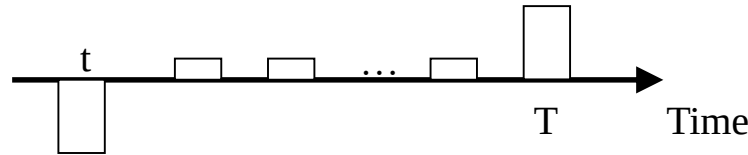
- Transaction with constant exposure:
 - A typical example for a constant EAD are zero-coupon-bonds:



- The *EAD* is represented by the notional value of the transaction over the whole maturity. Zero-coupon-bonds represent the simplest form of transactions for EAD estimation purposes.
- This transaction type is of minor importance for banks.
- However, it is still an important case to consider as many portfolio models will assume that all transactions can be represented by zero-coupon bonds.

EAD – Transaction Types

- Transactions with deterministic exposure:
 - The EAD is not constant over time, but we know upfront, the value of the EAD for each future point in time:



- Typical examples are exposures with fixed interest rate payments or annuities: term loans, mortgages, bonds, ...
- The precise value of the EAD depends on the legal possibility to demand future interest rate payments at default.

- Transaction types with stochastic exposure:
 - Commitments and lines of credit:
 - The obligor has a line of credit (e.g. an overdraft facility). She/he can use this line of credit, but she/he doesn't have to.
 - The EAD is dependent on the decision of the obligor.
 - Example: The current account overdraft facilities for a retail customer, e.g. the customer can draw 2,000 EUR. Then, the EAD can be between 0 and 2,000 in all future points in time.
 - It has to be considered, of course, that the usage of such lines will typically increase before a default.

- The exposure is dependent on conditions:
 - The exposure depends on the probability that certain events materialize.
 - Examples are more complex contracts for realizing projects or contracts linked to import/export activity.
 - E.g., loans for more complex building projects dependent on the completion of certain building phases.
 - Example: A building company receives a 10 million EUR loan. The loan agreement is structured in two tranches of 5 million EUR each. The first tranche is immediately available. The second tranche depends on the completion of the first building phases.
 - Thus, the exposure after 1 year is either 5 or 10 million depending whether the conditions in the contract are fulfilled.

- The exposure is dependent on market prices
 - The *EAD* is determined by market risk factors (e.g. interest rates).
 - This transactions type is mainly represented by the trading activity exposing the bank to counterparty risk in its OTC-activities (e.g. interest rate swaps).
 - When market risk factors develop in favor of the bank the OTC contract gains in value and an exposure with the counterparty (mainly other banks) opens up.
 - **Example:** A bank enters into a Payer Swap with another bank (counterparty). The swap has a notional value of 10 million EUR and the swap rate is 5%. On the next reset date the maturity is 3 years. The floating rate just before the reset date is also 5%.

Estimate the exposure on the reset date in case that all relevant interest rates are changing to 4% (Case 1) or 6% (Case 2)?

EAD – Transaction Types

Solution:

The value of the interest rate swap depends on the market value of the individual payments. The exposure is represented by the sum of these values..

The value of the floating payments is 100% or 10 million EU at the reset date by definition.

Case 1: Market value of the fixed payments:
$$= \frac{500,000}{(1.04)} + \frac{500,000}{(1.04)^2} + \frac{10,500,000}{(1.04)^3} = 10,277,509$$

Value for the bank = $10,000,000 - 10,277,509.1 = -277,509 \rightarrow \text{Exposure} = 0$

Case 2: Market value of the fixed payments:
$$= \frac{500,000}{(1.06)} + \frac{500,000}{(1.06)^2} + \frac{10,500,000}{(1.06)^3} = 9,732,698$$

Value for the bank = $10,000,000 - 9,732,698.8 = 267,301 \rightarrow \text{Exposure} = 267,301.2$

EAD – Estimation of the expected EAD

- Similar to the probability of default the estimate for the EAD depends on the time horizon (e.g. 1 year).
- All information concerning the exposure and potential changes have to be considered when estimating the EAD.
- In the simplest case (under the Basel regulation) all information is represented by one EAD for each obligor/obligation, representing an estimate of the EAD in the defined time horizon.

EAD – Estimation of the expected EAD

- Under the Basel regulation one number for the EAD is used in the calculation of the capital requirement.
- Problems and uncertainties in the estimation are addressed by using a conservative estimate.
- For transactions with constant or deterministic exposure the EAD is represented by the amount outstanding of the credit exposure. This is normally represented by the book value of the exposure (see e.g. Regulation (EU) 575/2013 Article 111). In most cases this represents a conservative estimate (e.g. consider an annuities).

EAD – Estimation of the expected EAD

- For undrawn lines of credit and commitments a credit conversion factor (CCF) is used. Values of 0%, 20%, 50% and 100% are applied. This depends on the maturity and the possibility to cut/revoke the line or commitment (Regulation 575/2013 Article 111).
- In the IRB Approach banks can provide own estimates of the CCF based on internal data.
- For transactions dependent on market values a conservative approach is applied. The starting point of the estimation is today's market value (replication value), if positive and zero otherwise. Furthermore, there is an add-on (as the market value could increase till default). This add-on is defined as a percentage of the nominal value and depends on the product type and the maturity.

Loss Given Default (LGD)

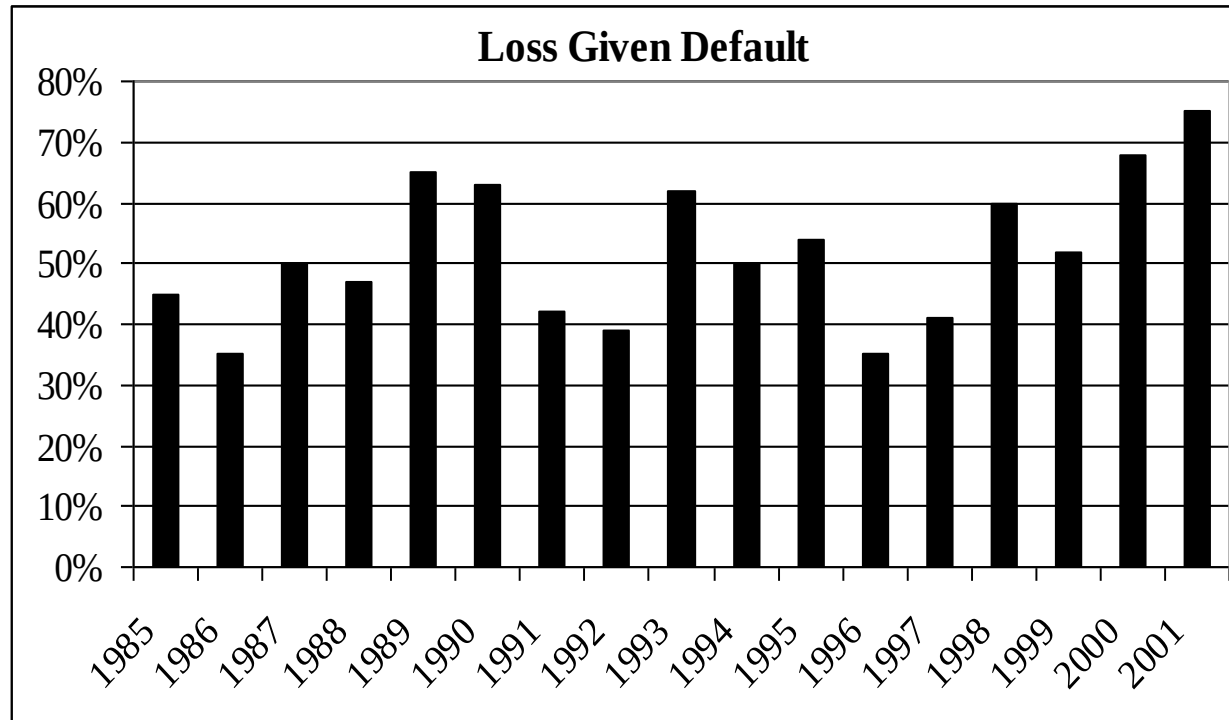
- Definition:

Loss Given Default (LGD): LGD is the fraction of the exposure that will be lost following default. Where “loss” means economic loss, including all relevant factors as discount effects, direct and indirect costs associated with collecting on the instrument.

Loss Given Default = $1 - \text{Recovery Rate}$

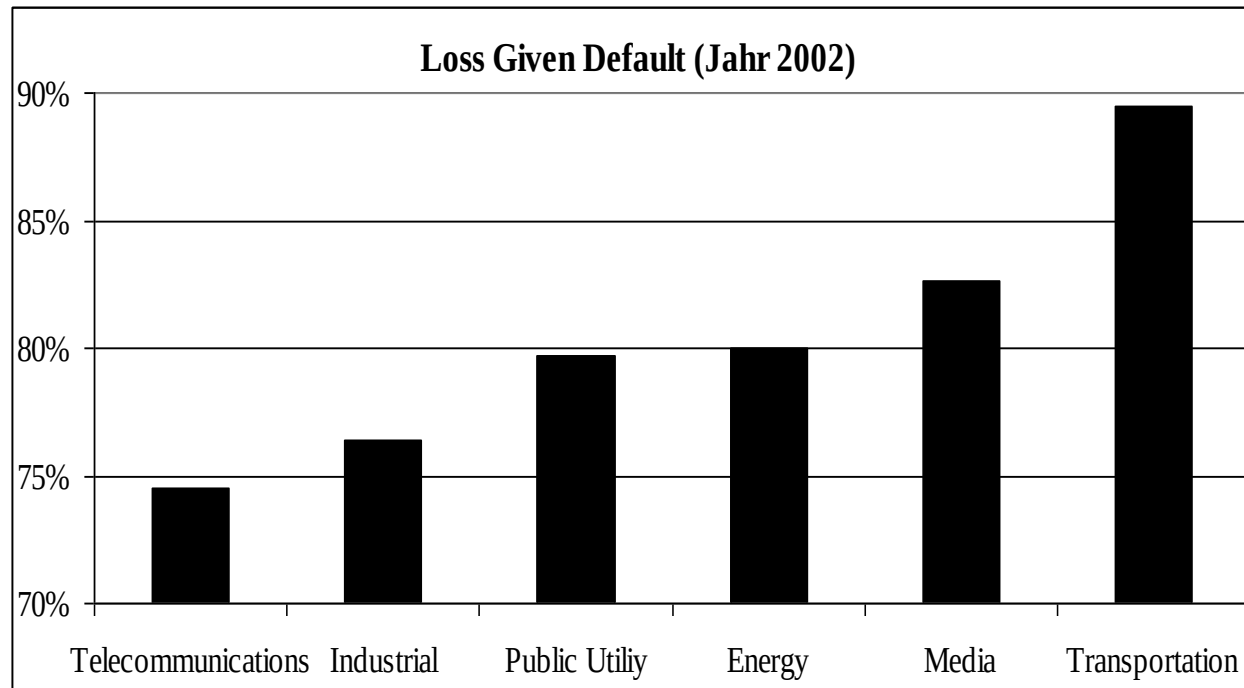
Introduction – Loss Given Default

- Time series of realized LGDs:



Introduction – Loss Given Default

- LGDs for different industries:



- LGD is an important parameter in the Basel II framework within the IRB approaches.
- Expected LGDs have to be estimated for all obligors in the advanced IRB approach.
- In contrast to the PD, in the IRB foundation approach expected LGD has not to be estimated by the credit institution, but is provided by the regulator:
 - Senior unsecured loans: 45%
 - Subordinated debt: 75%
 - Mortgage based debt: 35%

- The calculation of expected LGDs consists of two steps:
 - For past defaults the realized LGD has to be calculated. This implies that the default is already completely processed!!!
 - Based on this data pool the expected LGD can be estimated. This is mainly done by defining homogenous pools and observing the average realized LGD in these pools.

Introduction – Loss Given Default

- Basically two methods exist to calculate the realized LGD for a past default:
 - Realized LGD based on the credit institution's own work out data.
 - Market prices for defaulted debt (not applicable for most obligors).

- Calculation based on credit institution's own work out data:
 - In most cases the obligor in default is handled by a special department (work out department, collection department).
 - This department is responsible to recover a high as possible fraction of the defaulted exposure. For this all necessary steps are undertaken (restructuring, liquidation, realizations from collateral ...).
 - Proceeds and costs are booked on the account of the defaulted obligor or are available in other systems.
 - This dataset is the basis for the calculation of the realized LGD.

Realized LGDs for Past Defaults

- Proceeds and costs are realized after the default. The realized LGD is measured for the default date. Therefore these components have to be discounted.
- Applied and academic research literature has no clear answer which interest rate should be used for discounting (none?; risk-free?; risk-adjusted?)
- In the literature proposed interest rates span from 0% to 20%.
- A theoretically correct interest rate does not exist.
- Often a interest rate representing a risky exposure is applied and recommended by supervisors.

Realized LGDs for Past Defaults

- The realized LGD is thus calculated by

$$\text{LGD} = 1 - \frac{(\text{Sum of present values of proceeds} - \text{Sum of present values of costs})}{\text{EAD}}$$

- The second part of the calculation simply represents the recovery value
- Note that LGDs above 100% and below 0% are possible.
- An important driver of the LGD will be proceeds from collateral (e.g., real estate, cars, ...).

- Alternative Method: LGD calculation based on market data
 - Some loans and bonds can be traded on the financial markets even when in default.
 - In these situations the realized LGD can be based on the discounted market price of the credit exposure.
 - Advantage:
 - Fast completion of the default (within one month)
 - Simple calculation of the LGD
 - No extensive data collection
 - No extra costs in the collection department

Realized LGDs for Past Defaults

- Disadvantage:
 - These markets are often illiquid
 - Traded prices have high risk premia
 - Only a small fraction of loans or bonds are traded

- Summary:

For banks concentrated on retail or SME corporate loans market prices for defaulted debt is no real alternative in the LGD calculation.

However, these markets are growing fast and therefore the possibility to trade defaulted debt will strongly increase in the future.

Expected Loss Given Default

- Calculation of the expected LGD by pooling
 - Based on the characteristics of the credit exposure homogenous pools or groups are defined, such that exposures of one group are supposed to have identical expected LGDs:
 - Potential characteristics:
 - Type of obligor (e.g. retail, corporate, government, ...)
 - Obligor rating (e.g. AAA, AA, A, ...)
 - Type of exposure (e.g. revolving loan, ,...)
 - Vintage
 - Industry
 - Income, Turn-over, ...

Expected Loss Given Default

- Simple example for homogenous group:

All corporate customers with a turn-over below 10 million euro in the telecommunication industry

- For each homogenous group the expected LGD can be calculated as the arithmetic average of the realized LGDs in this group.
- Important to note: the number of groups has to be small enough to guarantee a sufficiently high number of observations per group.
- This method of LGD calculation is the simplest approach.

Expected Loss Given Default

- Example: Calculate the expected LGD for all possible groups:

Default	Type of exposure	Rating	Realized LGD
1	retail loan	A	40%
2	commercial loan	B	48%
3	retail loan	B	60%
4	retail loan	A	30%
5	commercial loan	A	36%
6	commercial loan	B	60%
7	retail loan	B	80%
8	commercial loan	A	25%
9	commercial loan	B	50%
10	retail loan	A	45%

Expected Loss Given Default

- Solution:
group 1: type of exposure: retail loan, rating: A
group 2: type of exposure: retail loan, rating: B
group 3: type of exposure: commercial loan, rating: A
group 4: type of exposure: commercial loan, rating: B

$$E[\text{LGD for group 1}] = (40\% + 30\% + 45\%) / 3 = 38.33\%$$

$$E[\text{LGD for group 2}] = (60\% + 80\%) / 2 = 70\%$$

$$E[\text{LGD for group 3}] = (36\% + 25\%) / 2 = 30.5\%$$

$$E[\text{LGD for group 4}] = (48\% + 60\% + 50\%) / 3 = 52.67\%$$

Potential extensions:

- Time series analysis:
 - The calculation of expected LGDs should be based on a time series of realized LGDs.
 - These time series allow to analyze trends or cyclical behavior (economic condition) in the data set.
 - This information can significantly improve the estimation of the LGD.
- Regression-based analysis:
 - The characteristics can be used in an regression approach similar as in the PD-estimation. This regression can also include economic data.
- In addition, the realized LGD data set can be used to estimate standard deviations or similar metrics to treat LGD as a random variable.

Credit Portfolio Models

- The credit risk measurement provides risk parameters (*PD*, *EAD* and *LGD*) for the individual obligors.
- To estimate the credit risk of the overall portfolio, it is necessary to aggregate all the information, to provide a risk estimate of the credit portfolio.
- Credit risk portfolio models allow to provide this estimation based on certain assumptions.
- Depending on the complexity of the model, the risk estimate can be represented by an analytical solution or by simulation methods.
- The mean difference across models is how concentration and correlation risks are treated.

- Standardized Approach:
 - The Standardized Approach of Pillar 1 represents one of the simplest methods to derive the credit risk of a portfolio.
 - This approach allows also to derive very simple estimations for the 99.9% quantile of the loss distribution under Pillar 2.
 - The parameterization of the Standardized Approach depends on the exposure class and the collateral.
 - The main focus is on the credit quality represented by external ratings in the Pillar 1. This logic can be used in the Pillar 2 based on internal ratings.

- Standardized Approach:

- The risk weight of an exposure for corporates can be determined using the following table (see Regulation (EU) 575/2013 Article 122).

Credit Quality Step:	1 (AAA, AA)	2 (A)	3+4 (BBB, BB, unrated)	5+6 (B, CCC)
Risk Weight (RW):	20%	50%	100%	150%

- The loss potential of the individual obligor is

$$RW \cdot EAD \cdot 8\%$$

- The loss potential for the portfolio represents the sum of the individual numbers.
- Thus, this methodology does not consider the concentration and correlation in the portfolio.

- In the previous approach the reason behind a default is not modelled at all. It has to be assumed the default is just an exogenous event with a certain probability.
- In the Merton Model the reason behind a default is explicitly modelled (structured model).
- In the model it is assumed that a corporation can be modelled by a stochastic asset value. A default is triggered when the asset value falls below a certain value.
- This default point is defined based on the leverage of the corporation.

- Merton [1974] models the asset value based on a geometric Brownian motion:

$$dV / V = \mu \cdot dt + \sigma \cdot dW$$

dV / V change in asset value (return)

dt infinitesimal time change

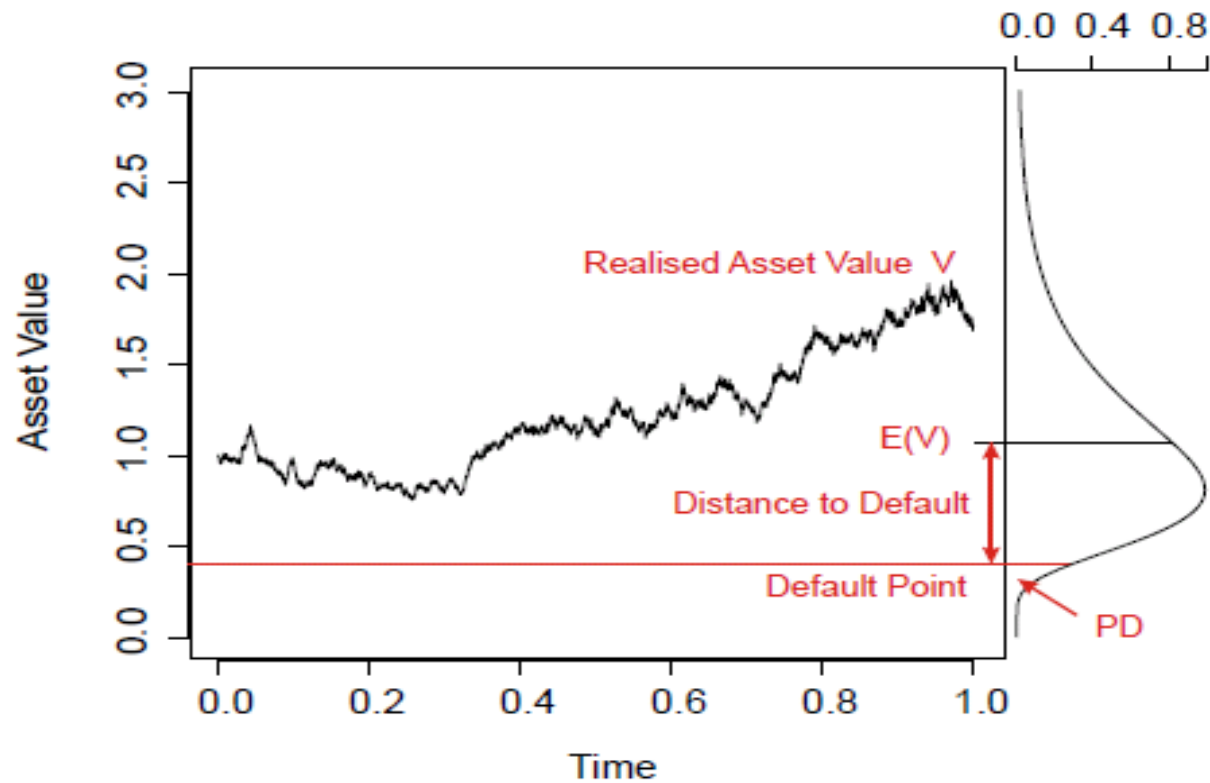
dW change in the stochastic process

μ expected return

σ volatility

- The asset value follows a log-normal distribution. Default occurs, if the asset value drops below the default point (D) at a certain point in time T .
- In the simplest case, D represents the overall debt and the point in time represents the maturity of this debt T .

Merton Models



- The asset value is modelled as:

$$V_T = V_0 \cdot \exp[(\mu - \sigma^2 / 2) \cdot T + \sigma \cdot W_T]$$

$$W_T \sim N(0; T)$$

- Based on this assumption $\ln V_T$ is normally distributed with

$$\ln V_T \sim N(\ln V_0 + (\mu - \sigma^2 / 2) \cdot T; \sigma^2 \cdot T)$$

- Based on this model, an important metric can be calculated: the Distance-to-Default (DD) defines by how many standard deviations the expected asset value has to decrease to reach the default point D at a certain time T :

$$DD = [\ln V_0 + (\mu - \sigma^2 / 2) \cdot T - \ln D] / (\sigma \cdot \sqrt{T})$$

- In addition, the standardized asset value can be defined as

$$Y_T = [\ln V_T - (\ln V_0 + (\mu - \sigma^2 / 2) \cdot T)] / (\sigma \cdot \sqrt{T}) \sim N(0, 1)$$

- Based on these definitions the probability of default can be estimated for the corporation using the following steps:

$$\begin{aligned}PD_T &= \text{Prob}(V_T \leq D) \\&= \text{Prob}(Y_T \leq -DD) \\&= N(-DD) = 1 - N(DD)\end{aligned}$$

This implies: $DD_i = -G(PD_i)$

$N(.)$ Distribution function of the standard normal

$G(.)$ Inverse of the distribution function of the standard normal

- The Merton Model is an example for a 1-factor model, as the stochastic asset value of the corporation is driven by one standard normal factor.
- This concept can be extended to multiple factor models. In these models the asset value is simply dependent on more than one factor. For example, let's consider a 2-factor model:

$$Y = w_1 \cdot X_1 + w_2 \cdot X_2$$

- These models allow to incorporate correlation between the defaults of corporations.

- The following model can be used to model the asset value of corporation i :

$$Y_i = w \cdot M + \sqrt{(1 - w^2)} \cdot \epsilon_i \qquad M, \epsilon_i \sim N(0, 1)$$

- This implies that the change in asset value is driven by a general market factor M (identical across all corporations) and by an idiosyncratic factor ϵ_i that only impacts corporation i .
- Note that Y_i is still standard normal and the weight w determined the correlation among corporations..
- The greater the weight of the market factor, the more likely are clustered defaults of corporations.

- IRB Advanced Approach:
 - The presented 2-factor model represents the starting point for the IRB Advanced Approach (see Regulation (EU) 575/2013 Article 153) to determine the capital requirement.
 - The inputs of the model are PD_i , LGD_i and EAD_i of obligor i .
 - The PD_i is used to determine DD_i of obligor i .
 - In addition a function is provided to determine the weight representing the correlation (denoted as R). The input into this function is PD_i implying that the correlation with the market factor is not identical across obligors.
 - Using the assumption that every loan in the portfolio is infinitely small (i.e. the fraction of the individual loan tends to zero), there is an analytic solution for the loss distribution and, thus, also an analytic solution for the 99.9% quantile.
 - This also implies that the risk contribution to the VaR (= capital requirement) of every individual obligor can be determined.

- IRB Advanced Approach:
 - For corporates the risk-weights can be estimated using

$$\text{Correlation (R)} = 0.12 \cdot \frac{1 - e^{-50 \cdot \text{PD}}}{1 - e^{-50}} + 0.24 \cdot \left(1 - \frac{1 - e^{-50 \cdot \text{PD}}}{1 - e^{-50}} \right)$$

$$\text{Maturity adjustment (b)} = (0.11852 - 0.05478 \cdot \ln(\text{PD}))^2$$

$$\text{RW} = 1.06 \cdot 12.5 \cdot \left[\text{LGD} \cdot N \left(\frac{G(\text{PD})}{\sqrt{1-R}} + \sqrt{\frac{R}{1-R}} \cdot G(0.999) \right) - \text{PD} \cdot \text{LGD} \right] \cdot \frac{1 + (M - 2.5) \cdot b}{1 - 1.5 \cdot b}$$

$G(.)$... Inverse of the standard normal distribution function

Loss potential: $\text{RW} \cdot \text{EAD} \cdot 8\%$

- IRB Advanced Approach:
 - The loss potential of the portfolio is represented by the sum of the individual loss potentials.
 - This model considers explicitly the correlation of obligors.
 - However, the concentration risk is not considered.
 - The precise estimation of the VaR for portfolios that have concentrations in individual names is not possible.
 - In general, the combination of correlation and concentration risk makes it impossible to analytically calculate the loss distribution.

- For models, that do not allow analytical calculations, the loss distribution can be simulated.
- This means that various simulation scenarios are generated and in each scenario the information is simulated whether a certain obligor defaults or not.
- Thus, in each scenario the loss in the portfolio can be calculated, i.e., one realization of the loss distribution is generated.
- Using a simulation with n (e.g. $n = 10,000$) scenarios allows to determine various metrics based on the simulated portfolio losses (e.g. expected value, standard deviation and quantiles)
- The same principles are used, as if a real sample is observed.

- Simulation models without correlation:
 - Using the presented idea, a 1-factor model can be simulated using the following steps:
 1. The DD_i of each obligor i is determined based on PD_i : $DD_i = -G(PD_i)$
 2. In each simulation scenario one standard normal random variable is Y_i generated, i.e. simulated.
 3. If this number is smaller than $-DD_i$, then the obligor i defaulted in this scenario.
 4. The loss of the defaulted obligor i is $EAD_i \cdot LGD_i$, otherwise it is 0.
 5. The loss of the portfolio in this simulation scenario is the sum of the individual losses.
 6. Steps 2 to 5 are repeated n times.

- Simulation models without correlation:
 - After generating many scenarios the observations can be used to calculate the 99.9% quantile.
 - The advantage of this model is that now concentration risks are fully considered.
 - This is achieved by simply using the individual EADs of the obligors.
 - In this simulation the default of one obligor is independent of the default of another obligor. Thus correlations are not considered.

- Simulation models with correlation:
 - The same logic can be applied for a 2-factor model. The only additional parameter that is needed is the weight w of the market factor:
 1. The DD_i of obligor i is determined based on PD_i : $DD_i = -G(PD_i)$
 2. In each simulation scenario, one standard normal random number is generated. This number represents the market factor M . Furthermore, one additional standard normal random number is generated for each obligor ϵ_i .
 3. For each obligor the standardized asset value is calculated: $w \cdot M + \sqrt{(1 - w^2)} \cdot \epsilon_i$
 4. If this number is smaller than $-DD_i$, then the obligor i defaulted in this scenario.
 5. The loss of the defaulted obligor i is $EAD_i \cdot LGD_i$, otherwise it is 0.
 6. The loss of the portfolio in this simulation scenario is the sum of the individual losses.
 7. Steps 2 to 6 are repeated n times.

- Simulation models with correlation:
 - After generating many scenarios the observations can be used to calculate the 99.9% quantile.
 - The advantage of this model is that now concentration and correlation risks are fully considered.
 - The complexity of these simulation models can be increased based on the focus of the model:
 - Use different weights for different obligors
 - Use more than two factors (e.g. one per industry)
 - Include simulations for LGD and EAD. For example, generate higher individual losses if in one scenario many defaults are observed.

Credit Risk Management

Introduction – Credit Risk Management

- The results of the credit risk estimation are an important input into various management tasks:
 - Risk-adjusted pricing
 - Risk-adjusted performance measurement
 - Limit systems
- In addition, the bank can actively manage the credit risk of the portfolio by issuing / trading credit derivatives:
 - Single-name credit derivatives, such as Credit Default Swaps (CDS)
 - Structured products, such as Asset-Backed Securities (ABS)
- These products offer the possibility to trade certain credit risks.

Credit Default Swaps

- Credit Default Swaps (CDS) are contracts to insure against losses due to the default of a certain reference entity (e.g. Bayer AG) for a certain notional amount and a certain period of time.
- The CDS (protection) buyer pays quarterly the CDS spread (e.g. 100 bp p.a.)
- In the case of default the CDS buyer can deliver defaulted debt (e.g. a bond) and the CDS seller has to pay the notional amount.

	No Default	Default
CDS Buyer	pays quarterly CDS spread e.g. 25 bp	delivers defaulted bond e.g. market price of 40%
CDS Seller	no payments 0	pays notional amount 100%

- How is the credit event defined?
 - Bankruptcy
 - Failure to Pay
 - Obligation Acceleration
 - Obligation Default
 - Repudiation/Moratorium
 - Restructuring
- Certain minimum notional amounts for the events (Default requirement: 10,000,000 USD / Payment requirement: 1,000,000 USD) and certain grace periods (e.g. 30 days) are defined.

- How are the deliverable reference assets defined?

Obligation:

- Payment (any obligation for payment of money)
- Borrowed Money (any obligation for payment of borrowed money)
- Reference Obligations Only
- Bond, Loan, Bond and Loan

Characteristics:

- Seniority: "Not Subordinated" or "Subordinated"
- Currency: specified or CAD/JPY/CHF/GBP/USD/EUR
- Others: Listing / Jurisdiction / Market of Obligations / Maturity (Restructuring)

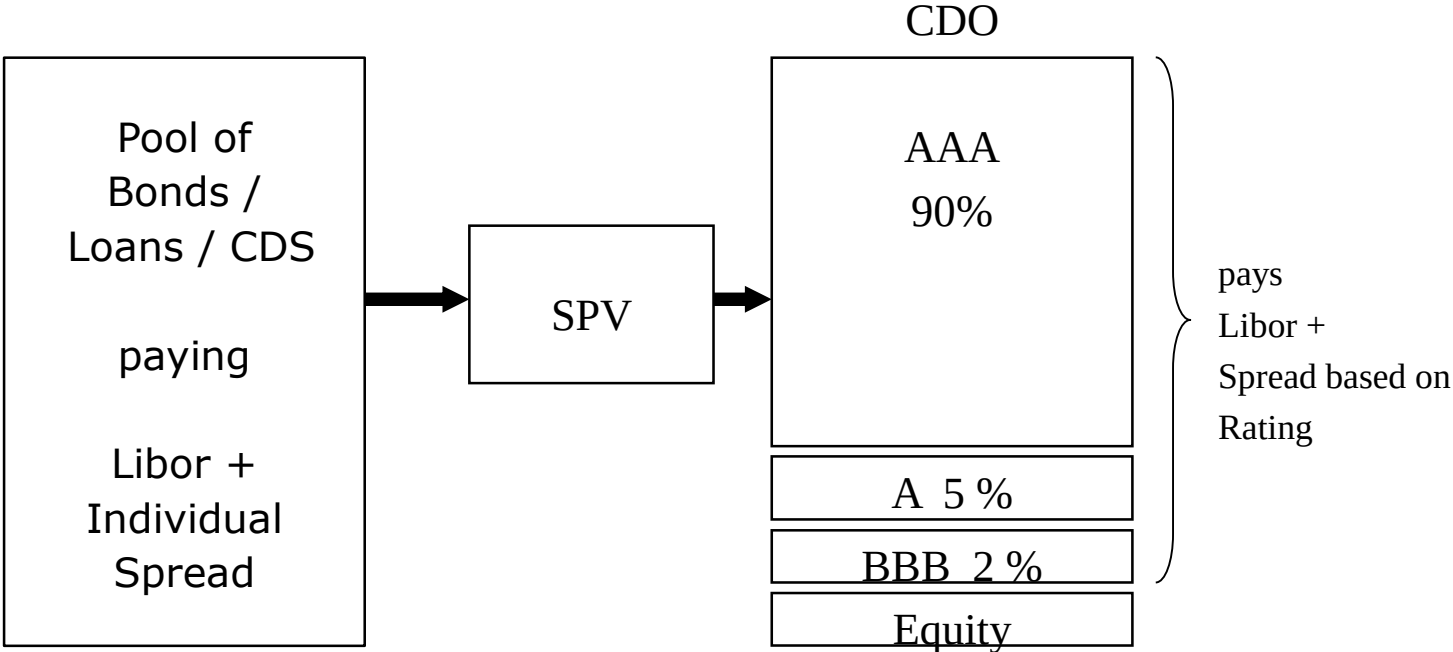
- Basically two forms of settlement are possible
 - Physical settlement
 - Cash settlement
- Cash settlement is by far the dominating form
 - After default the net CDS positions are determined.
 - Mostly auctions are used to determine a fair value.
 - The outcome is used to settle contracts.

- A CDS is an unfunded credit risk instrument, i.e., the market value of the product is zero at initiation and no payments are exchanged at initiation.
- Financial information systems (Reuters, Bloomberg) provide quotations for the CDS spread (similar to the swap rate in an interest rate swap).
- CDS are traded OTC.
- The number of entities, for which CDS are available, is continuously increasing. Indices based on CDS are available (i-Traxx or CDX family).

- The credit risk in the portfolio results from granting loans and buying bonds. The overall credit risk position of the bank could be far away from the optimum due to several unwanted effects:
 - Few obligors have a high overall credit exposure.
 - Obligors are clustered in similar industries.
 - The portfolio is „too“ diversified.
- CDS can be used to adapt the risk profile:
 - Hedging: Sell (part of) the credit risk of obligors leading to clusters.
 - Diversification: Buy credit risk of new obligors without direct business connection.

- Beside single-name credit derivatives, the trading in products based on credit portfolios is possible and increased a lot before the financial crisis. This portfolio products allow to trade the credit risk of small obligors.
- The trading in these products often creates synthetic risk positions in portfolios.
- The credit portfolios are either structure in one tranche (pass-through) or in multiple tranches with different credit risk. Examples are
 - Asset Backed Securities (ABS)
 - Mortgage Backed Securities (MBS)
 - Collateralized Debt Obligations (CDO)
- In the US, there still exists a very liquid market for structured products. Many of these products are now pass-through securities.

Structured Credit Risk Products



- These securities allow to sell certain parts of the portfolio that would lead to a credit risk concentration.
- On the other side, securities can be bought that allow to diversify the own portfolio.
- There are also products traded based on CDS indices that trade in a liquid market and allow to actively trade portfolio credit risk.
- An important critique regarding these products is that the pricing is complex and heavily depends on the correlation assumptions in the pool. Estimation errors can have significant non-linear effects because of the complexity of the tranches.
- These evaluation errors in the MBS in the US were the starting point of the financial crisis.

ICAAP Analysis

- The starting point of an analysis of the ICAAP process is the internal documentation of the bank and reports used for various purposes.
- These documents should be comprehensible for third party experts and should contain the following parts:
 - Business strategy
 - Risk strategy
 - ICAAP documentation
 - Documentation covering special topics: e.g. rating models, ALM, market risk model, ...
 - Stress testing concept
 - Internal and regulatory risk reports
 - Documentation regarding the organization and risk processes
 - Regulatory ICAAP questionnaire

- The documentation concerning the *Business Strategy* should contain a general mission statement, business goals, business areas, general structure of the organization and processes.
- The *Risk Strategy* should contain the following points:
 - Risk policy principles
 - Risk appetite
 - Risk structure (actual/target)
 - Risk Management (actual/target)
- The *ICAAP Documentation* is the central document describing the ICAAP process and should provide all the details regarding the various steps within the ICAAP.

- Documentations covering special topics should contain all the technical details concerning the estimation/quantification in the ICAAP.
- The *Stress Testing Concept* should present the process to define stress tests and the scenarios used in these tests.
- The risk reports document the results of the ICAAP calculations. These results can be compared with Pillar 1 results.
- Documentation regarding the organization and risk processes contains the setup of the ICAAP within the institution.
- Since 2009 the *ICAAP-Questionnaire* is sent out by the regulator to the banks asking to provide essential information concerning the ICAAP process.

ICAAP Analysis: Individual Steps

- To analyze the ICAAP process, the individual steps have to be evaluated based on the provided documents and questions/discussions with employees.
- Each step has to be represented by a reasonable implementation and documentation (considering the proportionality principle).
- The challenge is to identify areas/steps of the ICAAP process that only exist on paper, to find weaknesses in the implementation, which result in situations where the economic risk of the bank is not properly represented in the risk reports and to identify missing documentation.

- The risk identification has to represent more than just being a summary of Pillar 1 risks.
- A structured approach to evaluate all essential risk is necessary within the bank.
- Starting point is the existing risk structure. However, additional considerations/analyses are necessary:
 - What risk positions can the bank take any time? (No FX exposure at the moment does not imply that a business unit cannot open up a position.)
 - Thus, is there a process/mechanism in place to guarantee that non-material risk stay insignificant? Is there a limit system?
 - Analyze the business strategy / future plans.
 - Discuss these issues with the management.

- The documentation needs to contain the list of risk types and the evaluation results of the risk identification process.
- The documentation needs to explain the reasons for these results and provide additional analyses to foster the decision.
- Sometime material risk are considered to be insignificant because the bank would like to avoid that a quantification method has to be implemented (e.g., concentration risk).
- Often a general overview concerning the business activities of the bank is sufficient to evaluate the risk identification step.

ICAAP Analysis: Risk Quantification

- Default risk, credit risk concentration and residual risk from CRM techniques will be significant and high for most of the banks. Since credit risk is there main business activity.
- The usage of the Standardized Approach (often $8\% * \text{notional}$) will not be appropriate for most banks.
- Thus, most banks need a model representing at least the level of sophistication of the IRB Foundation Approach and need additional risk quantification for concentration and collateral risk (and potential feedback effects with market risks, e.g., for FX loans).
- Therefore, rating models need to be in place measuring PD and, for more advanced banks, LGD.
- These models represent essential elements for the validity of the ICAAP and need to be analyzed and validated.

ICAAP Analysis: Risk Quantification

- Many banks use standardized approaches for market, operational and interest rate risk, as these risk types are often considered to be low to medium.
- It is necessary to evaluate whether the bank documents that these risks are of minor importance and that the standardized approaches lead to conservative risk estimates.
- If complex securities are traded, then standardized approaches are not sufficient in many cases.
- Liquidity risk is mainly considered within the ILAAP process. Within the ICAAP the risk of an increase of the refinancing costs are considered.
- Other risks are often considered using capital buffers or not considered at all.

ICAAP Analysis: Risk-Bearing Capacity

- In most of the cases, risk aggregation is implemented by a simple summation of the individual results and, thus, is conservative in most circumstances.
- Concerning the risk-bearing capacity, the starting point in most cases is the regulatory capitals and, then, adjustments are applied.
- When analyzing the economic capital the treatment of hidden reserves is especially interesting, as these are often not considered in the risk estimation.
- In addition, the consideration of future profits has to be critically analyzed.
- Most problems exist when defining the capital buffer in the normative perspective.

ICAAP Analysis: Risk-Bearing Capacity

- The analysis of the risk-bearing capacity is the comparison of the risks with the available capital.
- This comparison has to be available for the bank-wide level and for the individual risk types.
- Often a link between capital limits and structural limits is missing. Being below the structural limit, often does not guarantee that the risk is also below the capital limits.
- Often limits are directed towards the usage, rather than being the outcome of risk-based management.
- These two circumstances hint at a situation, where the ICAAP process is only used for regulatory purposes and not for business decisions.

- Risk-related processes have two important dimensions:
 - The processes should clearly document how the ICAAP calculations work, including the stress tests.
 - In addition, it should be clear which risk management processes use the ICAAP results as input (reporting, pricing, performance measurement, ...).
- In case, it is not clear which processes incorporate ICAAP results, then it is likely that the ICAAP exists only isolated for regulatory purposes.
- It is important to have a kind of “use test”, as an isolated ICAAP process is often characterized by low quality.

- The ICAAP should be the responsibility of the risk management unit.
- This unit has to be independent from the business units, otherwise there will be feedback effects.
- The most relevant points are:
 - Independence up to and including the board level.
 - No business unit tasks and no profit-oriented compensation.
 - Enough resources.



VIENNA UNIVERSITY OF
ECONOMICS AND BUSINESS

**DEPARTMENT Finance, Accounting and
Statistics**

Welthandelsplatz 1
1020 Vienna, Austria

UNIV.-PROF. DR. RAINER JANKOWITSCH

T +43-1-313 36-4340
F +43-1-313 36-90-4340
rainer.jankowitsch@wu.ac.at
www.wu.ac.at

- Bhattacharya, S., Boot, A. W. A., and A. V. Thakor, 1998. "The Economics of Bank Regulation", *Journal of Money, Credit, and Banking*, 30, 745-770.
- Diamond, D. W., 1984. "Financial Intermediation and Delegated Monitoring", *Review of Financial Studies*, 51, 393-414.
- Hornik, K., Jankowitsch, R., Lingo, M., Pichler, S., and G. Winkler, 2007, "Validation of credit rating systems using multi-rater information", *Journal of Credit Risk*, 3, 1-27.
- Hornik, K., Jankowitsch, R., Lingo, M., Pichler, S., and G. Winkler, 2010, "Determinants of Heterogeneity in European Credit Ratings", *Financial Markets and Portfolio Management*, 24, 271-287.
- Lingo, M., and G. Winkler, 2008, "Discriminatory power: an obsolete validation criterion?", *Journal of Risk Model Validation*, 4, 4-24.
- Santomero, A. M., 1984. "Modeling the Banking Firm – A Survey", *Journal of Money, Credit, and Banking*, 16, 576-712.